

JGR Solid Earth

RESEARCH ARTICLE

10.1029/2025JB032010

Key Points:

- We compile three-decade long InSAR and GNSS time series spanning a series of VEI 4–5 eruptions in the Southern Andes
- Only two volcanoes in subduction zones show a clear pattern of pre-eruptive uplift, co-eruptive subsidence and post-eruptive uplift
- InSAR and GNSS geodetic data have recorded only the last pulses of pre-eruptive reservoir pressurization in many subduction zone volcanoes

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

F. Delgado,
fdelgado@uchile.cl

Citation:

Delgado, F., León-Ibáñez, P., Báez, J. C., & Parra, S. (2025). The triggering of sub-Plinian and Plinian eruptions: Insights from three decades of InSAR observations in the Andean Southern Volcanic Zone (1991–2025) and a global compilation. *Journal of Geophysical Research: Solid Earth*, 130, e2025JB032010. <https://doi.org/10.1029/2025JB032010>

Received 15 MAY 2025

Accepted 4 DEC 2025

Author Contributions:

Conceptualization: Francisco Delgado
Data curation: Francisco Delgado
Formal analysis: Francisco Delgado, Pablo León-Ibáñez, Juan Carlos Báez, Sofía Parra
Funding acquisition: Francisco Delgado
Investigation: Francisco Delgado
Methodology: Francisco Delgado
Project administration: Francisco Delgado
Resources: Francisco Delgado
Software: Francisco Delgado
Supervision: Francisco Delgado
Validation: Francisco Delgado
Visualization: Francisco Delgado
Writing – original draft: Francisco Delgado
Writing – review & editing: Francisco Delgado

© 2025. American Geophysical Union. All Rights Reserved.

The Triggering of Sub-Plinian and Plinian Eruptions: Insights From Three Decades of InSAR Observations in the Andean Southern Volcanic Zone (1991–2025) and a Global Compilation

Francisco Delgado¹ , Pablo León-Ibáñez¹ , Juan Carlos Báez² , and Sofía Parra¹

¹Department of Geology, Universidad de Chile, Santiago, Chile, ²Centro Sismológico Nacional, Universidad de Chile, Santiago, Chile

Abstract Volcanic eruptions are currently understood as triggered by increases in the overpressure of the magma storage regions, usually resulting in ground deformation. We assess whether observations of ground deformation can provide insights on the triggering mechanisms of sub-Plinian and Plinian eruptions, which are the most hazardous eruptions with a global impact. We process and compile InSAR and GNSS time series from 1991 to 2025 spanning four eruptions of this type in the Southern Andes: Hudson 1991, Chaitén 2008–2009, Cordon Caulle 2011–2012 and Calbuco 2015, resulting in the subduction zone with the largest erupted mass during 1980–2019. Only Cordon Caulle displays the theoretical pattern of pre-eruptive uplift due to reservoir pressurization, co-eruptive subsidence due to magma extraction, and post-eruptive uplift in response to magma inflow. For the rest of the volcanoes, we observe co- or post-eruptive ground deformation, but the data temporal resolution is low and did not sample well all the eruptions. On a global scale, InSAR and GNSS data recorded at volcanoes in subduction zones that experienced similar eruptions indicate few well-recorded eruptions. Only Okmok records a similar pattern to that of Cordon Caulle, while in other volcanoes geodetic data recorded the last episodes of pre-eruptive reservoir pressurization. This implies that most of the magma that increased the overpressure could have been emplaced in the decades or centuries prior to eruption and geodetic observations, or emplaced undetected. The longer time scales of recharge in subduction zone volcanoes compared with basaltic volcanoes in hot spots could indicate larger storage regions.

Plain Language Summary One of the grand challenges in volcanology is to forecast the onset and size of eruptions. These are triggered by increases in the magma overpressure until a threshold is reached triggering a dike intrusion and potentially an eruption. The magma overpressure usually results in measurable ground deformation before eruptions by satellite geodetic techniques such as InSAR. Here we compile InSAR time series in the Andean Southern Volcanic Zone before large explosive eruptions and compare them with a global compilation. We find that only a few volcanoes had suitable data for tracking pre-eruptive ground deformation, and that most of the magma that was erupted in these eruptions could have been accumulated many years before the eruptions.

1. Introduction

Current models for closed shallow reservoirs indicate that volcanic eruptions are triggered by a complex interplay of physical and chemical processes produced when magma is injected into the shallow crust in a storage region. This results in pressurization of the reservoir walls, until the tangential stress reaches the tensile strength of the rock, triggering a dike intrusion (Caricchi et al., 2021; Degruyter & Huber, 2014; Tait et al., 1989; Townsend, 2022). This rupture criterion is modulated by the reservoir size, rheological state, volatile content, and external stresses (Sigmundsson et al., 2020; Townsend, 2022). Other mechanisms that have traditionally been invoked as viable for triggering an eruption such as volatile exsolution (Tait et al., 1989) or changes in external stresses (Pinel et al., 2007) can trigger an eruption only if the magma storage region is close to failure, so their role is secondary (Annen et al., 2023; Townsend, 2022), at least on the short time scales of geophysical monitoring. The storage and transport of these magma batches can result in both seismicity and ground deformation on the Earth's surface that can be tracked using ground and space-borne geodetic methods such as tiltmeters, GNSS, and InSAR (Acocella et al., 2023; Lu & Dzurisin, 2014; Pinel et al., 2014; Sigmundsson et al., 2018).

In the past two decades advances in volcano monitoring, and several well monitored eruptions that have erupted 0.01–1 km³ of basaltic lava in Iceland (Eyjafjallajökull 2010, Bárðarbunga 2014, Fagradalsfjall 2021, Sundhnúkur 2023–2024) (Sigmundsson et al., 2010, 2020, 2022, 2024; Gudmundsson et al., 2016; Parks et al., 2025), and Kilauea (Anderson et al., 2019, 2023; Patrick et al., 2020; Roman & Lundgren, 2021, 2023; Segall et al., 2022), Sierra Negra (Bell et al., 2021a, 2021b; Gregg et al., 2022; Shreve & Delgado, 2023), Ambrym (Shreve et al., 2019, 2022), Fani Maore (Feuillet et al., 2021), and Nyiragongo (Smittarello et al., 2022) volcanoes have yielded a large amount of data that have allowed putting bounds on the dimensions, and storage conditions of levels of magma, and the triggering mechanisms. In several of these eruptions (Bárðarbunga, Kilauea, Sierra Negra, Ambrym, Fani Maore, Nyiragongo), magma was transported along kilometer-long dikes away from the storage regions, resulting in caldera collapse for Bárðarbunga and Kilauea (Anderson et al., 2019; Gudmundsson et al., 2016), incipient collapse for Sierra Negra (Shreve & Delgado, 2023), and drainage of summit lava lakes for Ambrym (Shreve et al., 2019), Kilauea (Neal et al., 2018) and Nyiragongo (Smittarello et al., 2022). For some volcanoes, intrusions of magma batches increased the magma overpressure until reaching the rupture point. This occurred in response to a nearly continuous magma supply (Kilauea 2018, Sierra Negra 2018, Gregg et al., 2022; Patrick et al., 2020), magma backup from multiple zones of storage (Kilauea 2018), discrete intrusions that accumulated for more than one decade before the eruption (Eyjafjallajökull 2010), or an increase in the deformation rate in the year before the eruption, possibly indicative of an increase in magma flux (2018 Sierra Negra, Bell, Hernandez, et al., 2021). In contrast, signs of pre-eruptive unrest were very small for some of these eruptions suggesting: (a) viscoelasticity relaxing the magma overpressure due to magma buoyancy before the 2014 Bárðarbunga eruption (Sigmundsson et al., 2020), (b) a release in tectonic stress, resulting in a decrease in seismicity and deformation just before the 2021 Fagradalsfjall eruption (Sigmundsson et al., 2022), (c) were indirectly evidenced by the end of degassing-induced depressurization one year before the 2018 Ambrym eruption (Shreve et al., 2022), (d) or were completely absent immediately before the 2021 Nyiragongo eruption (Smittarello et al., 2022). The Eyjafjallajökull eruption is an intermediate case because reservoir pressurization due to magma injection triggered a small basaltic lava flow followed by a sub Plinian andesitic eruption.

These events and older eruptions like the 1975–1984 Krafla Fires led Galetto et al. (2022) to propose that eruptions in basaltic calderas are triggered when the magma flux is higher than 0.1 km³/yr (3.1 m³/s). High fluxes increase the strain rate, keep the crust brittle around storage regions, resulting in eruptions in less than 1 year when the flux is sustained over time (Galetto et al., 2022). For example, the volume flux before the 2023–2024 Grindavik dike intrusion and Svartsengi eruptions reached 0.075–0.28 km³/yr (2.4–9 m³/s) (Li et al., 2025; Parks et al., 2025; Sigmundsson et al., 2024), significantly larger than the threshold of 3.1 m³/s. Pre-eruptive ground deformation also tends to follow an exponential trend indicative of mass injection into shallow storage regions (Pinel et al., 2010). Pre-eruptive models also exist for smaller Strombolian basaltic eruptions with a VEI up to 2, in which tiltmeter data show pre-eruptive uplift occurring only a few minutes before the eruption (e.g., Ripepe et al., 2021).

A significant caveat of these observations and models is that almost all of them were derived from volcanoes that erupted in either hot spot or divergent plate boundaries, except for Ambrym. This hampers the generalization of criteria derived from these eruptions to volcanoes in subduction zones, which are the regions with the largest amount of sub-aerial erupted mass in the past four decades (Galetto et al., 2023). This results from the explosivity of these eruptions, a consequence of the larger amount of water vapor that can be dissolved in intermediate and silicic magmas that usually erupt in these tectonics settings, resulting in large explosive Plinian and sub Plinian eruptions (e.g., Eichelberger, 1995; Gonnermann, 2015), and from other factors like larger zones of magma storage (Townsend et al., 2019). The protracted growth of larger reservoirs, particularly in silicic calderas, usually results in much longer pressurization time scales and eruption intervals (Townsend et al., 2019). Therefore, these eruptions can be significantly larger, more explosive, and more dangerous than those from hot-spots and divergent margins. Several tens of eruptions of that size have occurred during the same time span (Table 1), but only a few of them were monitored in detail to attempt analyses of the same type than those for basaltic eruptions (e.g., Cordón Caille 2011–2012, Basualto et al., 2023; Delgado, 2021). For example, the 2022 Hunga Tonga–Hunga Ha'apai submarine eruption is the largest eruption of the 21st century so far, and was not monitored in the near field.

In this study, we focus on measuring ground deformation in volcanoes that have experienced sub-Plinian and Plinian eruptions with volcanic explosivity indexes (VEI) of 4 and 5 since 1991, which is the date on which

Table 1

Summary of Observations of Ground Deformation at Volcanoes That Experienced VEI > 4 Eruptions During 1991–2025 (Figure 1), Regardless of the Magma Composition

Volcano	Eruption year	VEI	Pre-eruptive	Co-eruptive	Post-eruptive	Reference
Pinatubo	1991	6	Y ^T	N/A	N/A	Ewert et al. (1996)
Hudson	1991	5	N/A	N/A	Y	Pritchard and Simons (2004a), Delgado et al. (2014); this study
Spurr	1992	4	N/A	N/A	Y ^G	Lu and Dzurisin (2014); https://avo.alaska.edu ; this study (Table S5 in Supporting Information S1)
Láscar	1993	4	N	N	N	Pritchard and Simons (2004b)
Rabaul	1994	4	Y ^L	N/A	N/A	McKee et al. (2018)
Ulawun	2000	4	N/A	N/A	N/A	Not studied with InSAR
Ruang	2002	4	N/A	N/A	N/A	Not studied with InSAR
Reventador	2002	4	N/A	N/A	N/A	Not studied with InSAR
Manam	2005	4	N/A	N/A	N/A	Not studied with InSAR
Shiveluch	2005	4	N/A	N/A	N/A	Not studied with InSAR
Rabaul	2006	4	Y ^G	Y ^G	Y	Garthwaite et al. (2019), Saunders et al. (2023)
Chaitén	2008	4	N	Y	N	Delgado et al. (2022); this study
Kasatochi	2008	4	N/A	N/A	N/A	Submarine eruption
Okmok	2008	4	Y	Y	Y	Biggs et al. (2010), Lu and Dzurisin (2010), Xue et al. (2020)
Sarychev	2009	4	N/A	N/A	N/A	Not studied with InSAR
<i>Eyjafjallajökull</i>	2010	4	Y	Y	Y ^G	Pedersen and Sigmundsson (2004, 2006), Sigmundsson et al. (2010), Hreinsdóttir (2019)
Merapi	2010	4	Y ^E	Y ^E	N	Chaussard et al. (2013), Surono et al. (2012)
Shiveluch	2010	4	N/A	N/A	N/A	Not studied with InSAR
<i>Grímsvötn</i>	2011	4	Y ^G	Y ^G	Y ^G	Sigmundsson et al. (2018)
<i>Nabro</i>	2011	4	N/A	Y	Y	Goitom et al. (2015), Hamlyn et al. (2014)
Cordón Caulle	2011	5	Y	Y	Y	Delgado (2021); this study
Kelud	2014	4	Y ^T	Y	N	Hidayati et al. (2019), Cassidy et al. (2019)
Sangeang Api	2014	4	N/A	N/A	N/A	Not studied with InSAR
Manam	2014	4	N/A	N/A	N/A	Not studied with InSAR
<i>Wolf</i>	2015	4	N/A	Y	Y	Xu et al. (2016)
Calbuco	2015	4	N	Y	N	Delgado et al. (2017); this study
Sinabung	2018	4	Y ^G	Y ^G	N/A	Kriswati et al. (2024)
Sinabung	2019	4	N/A	N	N/A	This study (Table S5 in Supporting Information S1)
Raikoke	2019	4	N/A	N/A	N/A	McKee et al. (2021a)
Ulawun	2019	4	N/A	N/A	N/A	McKee et al. (2021b)
Taal	2020	4	Y	Y	Y	Galgana et al. (2014), Rivera et al. (2019), Bato et al. (2021)
Soufrière St. Vincent	2021	4	Y	Y	N	Camejo-Harry et al. (2024)
Semeru	2021	4	N	N	N	Galetto et al. (2025), this study (Table S5 in Supporting Information S1)
Fukutoku-Oka-no-Ba	2021	4	N/A	N/A	N/A	Submarine eruption
Hunga Tonga-Hunga Ha'apai	2022	5	N/A	N/A	N/A	Submarine eruption
Shiveluch	2023	4	N	N/A	Y	Volkova et al. (2024), this study (Figure S13 and Table S5 in Supporting Information S1)
Ruang	2024	4	N/A	N/A	N/A	Not studied with InSAR

Note. Volcanoes of the SVZ are shown with names in bold and volcanoes in hot spots and continental rift zones are shown with names in italic. The VEI magnitudes are taken from the Global Volcanism Program (<https://volcano.si.edu>), except for Okmok. Y indicates deformation detected by InSAR, Y with a superscript indicates deformation detected by ground geodetic techniques (T for tiltmeter, G for GNSS, E for electronic distance meters, and L for leveling), N to no deformation detected by InSAR, and N/A is not applicable due to a variety of factors. These include lack of data (Pinatubo, Hudson, Reventador, Nabro), networks destroyed by eruptions (Pinatubo, Merapi), volcanoes seasonally or permanently covered by ice and snow (Spurr, Grímsvötn, Shiveluch), volcanoes covered by tropical rainforest (Ulawun, Rabaul, Manam, Reventador, Sinabung, Ruang, Semeru), volcanoes in small islands not suitable for InSAR (Kasatochi, Sarychev, Raikoke, Manam, Sangeang Api), submarine volcanoes, or eruptions that to our knowledge have not been studied with InSAR. Semeru volcano subsided during 2014–2023, but this signal is localized on one small region of the volcano and not related to the 2021 VEI 4 eruption (Galetto et al., 2025).

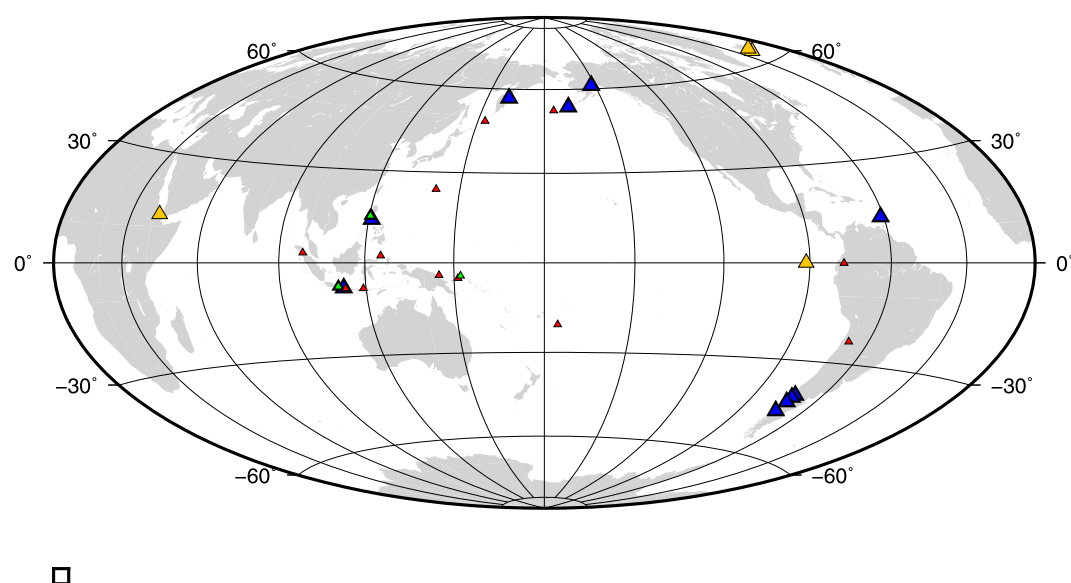


Figure 1. Volcanoes in subduction zones that experienced a VEI > 4 eruption during 1991–2025 (triangles, Table 1). Blue triangles indicate volcanoes where ground deformation was detected by InSAR before, during and/or after eruptions. Green triangles show observations of ground deformation made by GNSS, tilt meters, leveling or EDM before, during and/or after eruptions. Red triangles are volcanoes where no detection was possible. Yellow triangles are hot spot and continental rift volcanoes that experienced a Plinian eruption during the same period (not analyzed in this study). Most of the subduction volcanoes where deformation was detected related to these eruptions are located in the Southern Andes.

InSAR data started to be acquired on a global scale. First, we assess whether ground deformation is related to reservoir pressurization in order to systematically explain the triggering mechanism of these eruptions on a global scale. Second, we assess if volcanoes that have experienced these eruptions follow similar ground deformation patterns or not compared with those from basaltic volcanoes in hot-spots and rifts (e.g., Galetto et al., 2022; Gregg et al., 2022). These are clearly simplified hypotheses, because other factors must also be considered to systematically understand the triggering of eruptions. Some of them that do have an effect are the tectonic regime that can influence episodes of rifting (e.g., Sigmundsson et al., 2024), megathrust earthquakes (e.g., Caricchi et al., 2021), the velocity and obliquity of the subducting plate, and the crustal thickness and rheology of the upper plate (Acocella, 2021). However, we do not consider them because geodetic data in volcanoes are mostly sensitive to deformation resulting from deviatoric stresses such as those from magma overpressure, while secular or transient signals resulting from large-scale tectonics are usually smaller and/or removed from the data when working at the scale of individual volcanoes. The lack of detection also implies that deformation could not be well recorded and/or detected due to a variety of reasons including deep reservoirs, viscoelastic relaxation, and low recharge rates that result in deformation below the instrument detection limit, among others. These cases are largely unconstrained, so we do not consider them.

To meet the two objectives, our study presents new data and a global compilation. We focus on the Southern Volcanic Zone (SVZ) of the Andes of Chile and Argentina, which is the most productive subduction arc during 1980–2019 along with the Kamchatka arc in Russia, with erupted masses similar to those of the Hawai'i hot spot (Galetto et al., 2023). The large mass extrusion in the SVZ results from one VEI 4–5 eruption per decade (1991 Hudson, 2008–2009 Chaitén, 2011–2012 Cordón Caulle, 2015 Calbuco). These volcanoes have a wide range of magma compositions from andesitic to rhyolitic and a high density of satellite geodetic data that recorded ground deformation related to these eruptions (Table 1 and Figure 1). We used multi-platform InSAR time series from the ERS-1/2, ENVISAT, ALOS, RADARSAT-2, TerraSAR-X, COSMO-SkyMed, Sentinel-1, ALOS-2, and SAOCOM-1 missions from 1993 to 2025 from new data and previous studies (Delgado, 2021; Delgado et al., 2017, 2022, 2024; Reath et al., 2019), complemented with GNSS data when available. The sample of only four volcanoes in the SVZ is statistically sparse, so we compare our geodetic observations in the SVZ with those of eruptions of similar magnitude elsewhere (Table 1). Not all the eruptions in the compilation have been studied with geodetic data, so we also process InSAR data spanning selected eruptions and volcanoes in the Aleutians, Kamchatka and Indonesia where L-band data are available (Figure 1 and Table 1). Previous compilations of

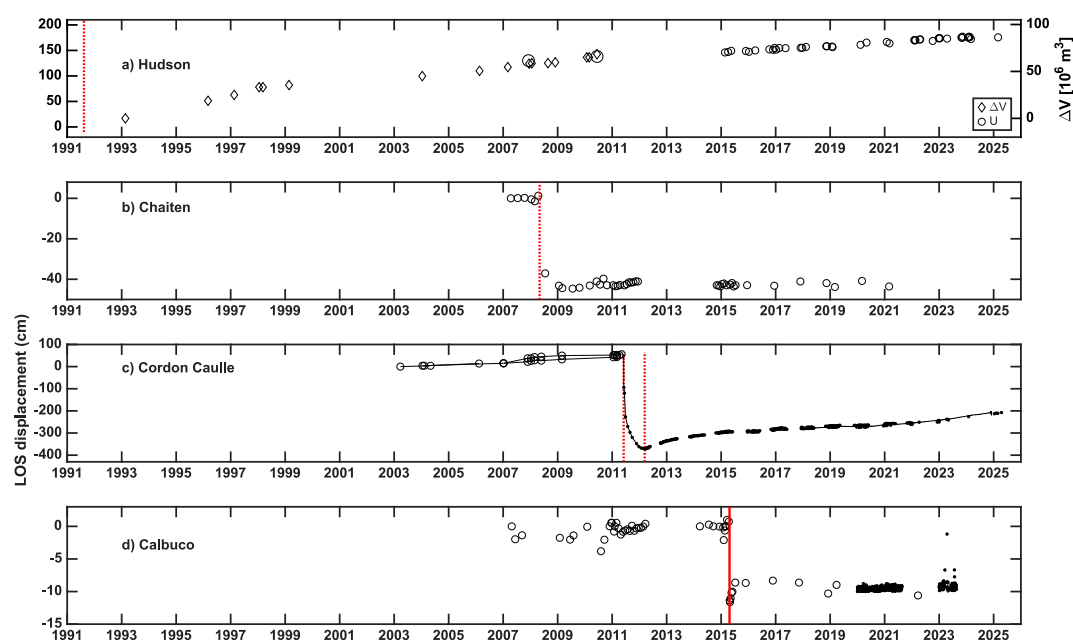


Figure 2. Time series of ground deformation at the four volcanoes in the SVZ that experienced VEI 4 or larger eruptions between 1991 and 2024. The scales are LOS displacement and vertical displacement for the GNSS time series (dots in panels b, c). Panel (a) shows a time series of volume change from 1993 to 2010 (right axis).

ground deformation data in volcanoes (Biggs et al., 2014, 2016; Biggs & Pritchard, 2017; Kilbride et al., 2016; Reath et al., 2019; Yip et al., 2022) did not address in detail the role of deformation due to reservoir pressurization and the size of the resulting eruption, except for Galetto et al. (2022) who studied the triggering of eruptions in basaltic calderas.

2. Results

Here we describe the ensemble of time series for the four VEI4–5 eruptions in the SVZ (Figure 2). Data processing and each data set are described in detail in the (Text S1–S3, Tables T1–T4 in Supporting Information S1), and here we present them with respect to each other. We note that they do not have the same temporal resolution and do not cover exactly the same time span. Ground geodetic monitoring data were nonexistent or were very limited in all of them.

The 1991 Hudson eruption was one of the largest of the 20th century, starting with a basaltic dike intrusion on August 8 and followed by an andesitic Plinian phase on August 12 (Kratzmann et al., 2009). The whole sequence lasted 1 week. The first ERS-1 SAR image ever over the Earth was acquired on 27 July 1991 (<https://www.esa.int>), 2 weeks before the eruption. However, no data for SAR were acquired during the eruption. InSAR data show nearly continuous uplift from 1993 to 2025 (Delgado et al., 2014; Pritchard & Simons, 2004a; Figure 2a), and InSAR time series spanning 2015–2025 record an average line-of-sight (LOS) ground velocity of ~ 3 cm/yr. The data can be modeled by a Mogi source located beneath the center of the caldera at depths of 7–10 km (Figures S1 and S2 in Supporting Information S1, Delgado et al., 2014). The source volume change has decreased from 0.023 km³/yr during 1993–1999 to 0.09 – 0.014 km³/yr during 2004–2025, and yields a total volume change of ~ 0.5 km³ during the 32 years covered by data. Because of the paucity of SAR data between missions, it is not clear whether the deformation data follow an exponentially decreasing trend or not, although (Delgado et al., 2014) suggested that the change in rate between 1993–1999 and 2004–2010 is consistent with a decrease in the pressure gradient driving the volcano uplift.

The 2008–2009 Chaitén eruption was the first rhyolitic eruption scientifically instrumented and observed in real time (Major & Lara, 2013), involving an explosive phase followed by a hybrid explosive-effusive phase with the extrusion of a dome. Like Hudson, there were no available SAR data that resulted in coherent interferograms before 2007 (Figure 2b). A time series spanning 1 year before the eruption shows no clear evidence for pre-

eruptive ground deformation above the background level of noise. During the eruption InSAR data recorded ~ 40 cm of LOS subsidence due to the closing of a dipping and a horizontal sill. The temporal sampling is less than one image per month, but during the first month and a half of the eruption the data show transient deflation lasting a few weeks only, and mostly absent during the effusive phase of the eruption (Delgado et al., 2022). No clear deformation indicative of post-eruptive recharge followed the eruption from 2009 until 2021 (Figure 2b and Figure S3 in Supporting Information S1).

The 2011–2012 Cordón Caulle eruption is the best scientifically instrumented and visually observed rhyolitic eruption to date (Castro et al., 2013; Delgado et al., 2019), and the one whose data have the highest temporal resolution available. It also started with an explosive phase followed by a hybrid explosive-effusive phase with the extrusion of a lava flow. Delgado (2021), presented a detailed study of ground deformation from 2003 to 2020, which we extend until 2025 (Figure 2c and Figure S4 in Supporting Information S1). The volcano rose ~ 0.6 m during 2003–2011, subsided 4.2–4.5 m during the eruption, and has risen ~ 1.6 m since the end of the eruption in March 2012 (Figure 3). The post-eruptive deformation can be explained by pressurization of either a horizontal sill at a depth of 6.2 km or a pressurized prolate spheroid at a depth of 4.5–5.2 km (Delgado et al., 2016, 2019; Novoa et al., 2022; Figures S5–S11 in Supporting Information S1). This likely results from the injection of basaltic magmas below the edifice, with a total volume change of 0.28 km^3 between March 2012 and April 2025, or a time-averaged volume flux of $0.021 \text{ km}^3/\text{yr}$ (Figure 4).

The 2015 basaltic-andesitic to andesitic Calbuco eruption involved two pulses. The first pulse lasted for 3 hours, followed 4 hours later by a second pulse that lasted 6 hours (Castruccio et al., 2016). The deformation trend is similar to that of Chaitén (Figure 2d). No signs of pre-eruptive uplift were recorded in the years before the eruption with InSAR, and up to 1.5 days before the eruption (Delgado et al., 2017). The volcano subsided during the eruption due to depressurization of a near vertical prolate spheroid source located at depths of ~ 8 – 10 km (Delgado et al., 2017; Nikkhoo et al., 2016). The time series from post-eruptive InSAR and GNSS data do not show clear ground deformation resulting from post-eruptive refill of the shallow reservoir (Figure S12 in Supporting Information S1).

3. Discussion

3.1. Eruption Triggers in the SVZ

Despite their large magnitude, few studies have addressed the triggering mechanisms of the cluster of four VEI4–5 eruptions in the SVZ from geological or geophysical data other than InSAR. Here we discuss them in detail.

The triggering mechanisms of the 1991 Hudson and 2008–2009 Chaitén eruptions have been poorly studied. For the Hudson eruption, Kratzmann et al. (2009) proposed a basaltic intrusion mechanism into an andesitic chamber in which the two magmas mixed with each other, and a certain amount of fractional crystallization occurred. This led to a dike intrusion that triggered the basaltic eruption, and a few days later the andesitic Plinian eruption. In contrast, for the Chaitén eruption the triggering mechanism has not been studied in detail yet.

The 2011–2012 Cordón Caulle eruption is by far the best studied event of the four eruptions here, but the triggering mechanism has also been little studied. Winslow et al. (2022) identified coarsely crystalline enclaves in the 2011 lava flow that are volumetrically small (1%) and could be interpreted as a potential trigger for magma injection. However, they lack disequilibrium textures indicative of interaction with magma of a different composition. Basualto et al. (2023) showed that VT seismicity slowly increased during the 3 months before the eruption, while the LP seismicity, indicative of magmatic fluids, increased in the days before the event. The ground deformation detected between 2003 and 2011 can then be regarded as one of the signs that an eruption would be triggered by an increase in the overpressure of the plumbing system (Delgado, 2021; Jay et al., 2014). Due to the increase in LP seismicity, we consider that the eruption was triggered by an increase in internal overpressure, with no need to argue for an external mechanism such as poorly located earthquakes in nearby faults (Novoa et al., 2022).

For the 2015 Calbuco eruption, two models explain the triggering of this eruption. Morgado et al. (2019) suggested heating of the crystal mush 4 days before the eruptions, while Arzilli et al. (2019) proposed a trigger based on second boiling and pressurization of the plumbing system. For the second phase, decompression exerted by the emptying of the reservoir in the first phase results in second boiling that triggered this eruption 3 hr later

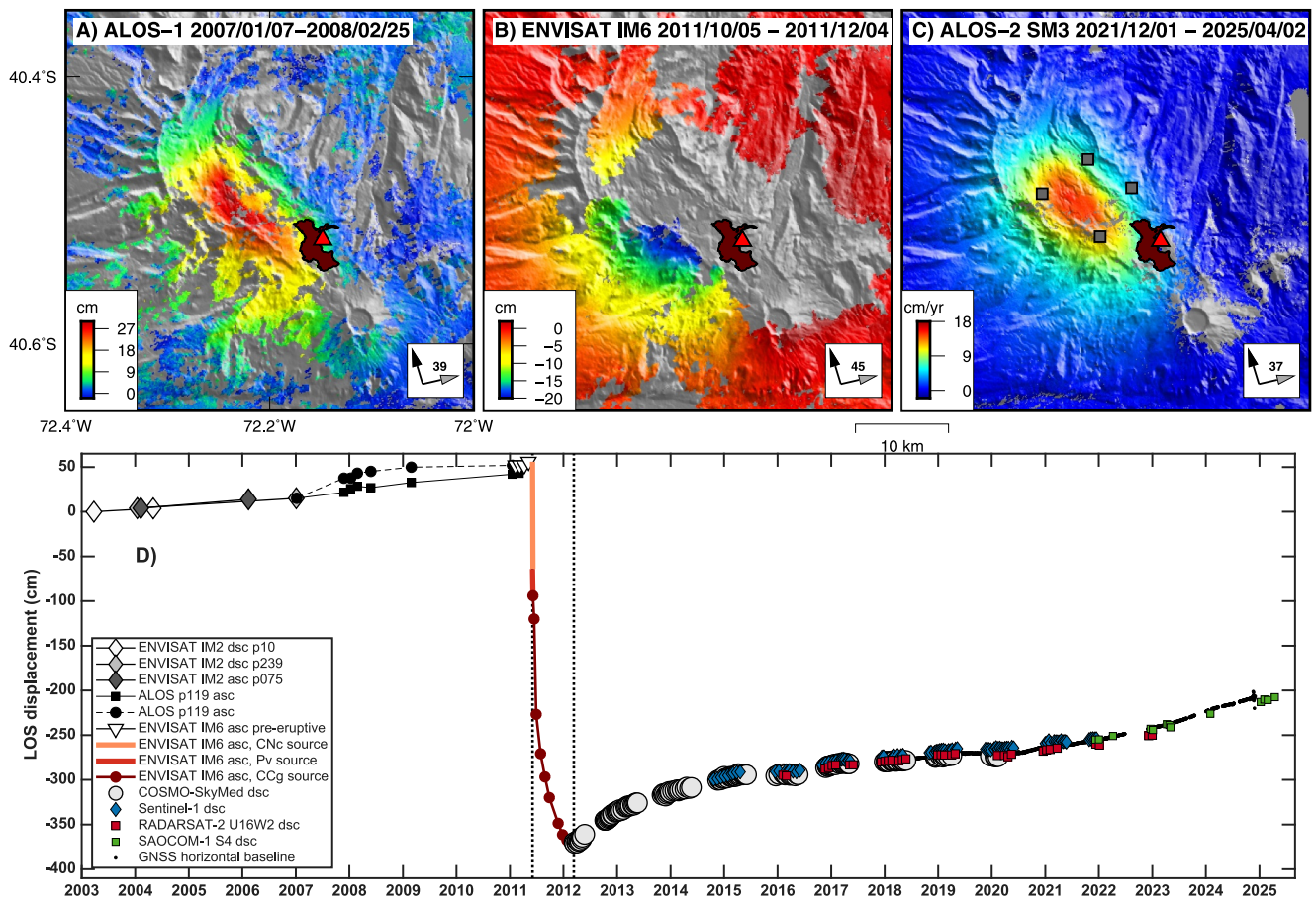


Figure 3. Cordon Caulle data. (a) Pre-eruptive ALOS ascending interferogram, (b) co-eruptive ENVISAT ascending interferogram, (c) mean ground velocity from post-eruptive ALOS-2 ascending InSAR time series. The gray squares show the location of the four GNSS monitoring stations in the volcano. The red triangle and the brown outline are the 2011 eruptive vent and lava flow. (d) Time series of ground deformation spanning 2003 to 2025 (zoomed-in from Figure 2c, updated from Delgado, 2021). They are manually shifted to lie on top of each other for the same pixels of maximum ground deformation. The GNSS horizontal baseline is calculated between the easternmost and westernmost stations. The data show 0.6 m of pre-eruptive uplift, 4.2 m of co-eruptive subsidence and 1.6 m of post-eruptive uplift. Asc and dsc refer to ascending and descending data. The ENVISAT co-eruptive time series are synthetic data calculated from three best-fit source models (CNc, Pv, CCg) and pressure changes (Delgado et al., 2019). The letter *p* followed by a number refers to satellite path and path number. IM2 and IM6 are the image mode 2 and 6 beams of the ENVISAT radar, S4 is SAOCOM-1 swath 4 and U12W2 is the Wide Ultra Fine 16 beam of RADARSAT-2.

(Castruccio et al., 2016). None of these mechanisms can be compared with models derived from InSAR data because pre-eruptive ground deformation was not detected, small intrusions with $\Delta V = 0.005 \text{ km}^3$ at depths of 10 km do not result in clear ground deformation signal and the low temporal resolution of the data (3 hr between the two pulses versus 12 days of every Sentinel-1 image).

From a geodetic point of view, we can classify these eruptions in two sets. First, volcanoes that show a near constant uplift after a sub Plinian or Plinian eruption for more than one decade, like Hudson and Cordón Caulle with rates of $\sim 3\text{--}4 \text{ cm/yr}$ and $15\text{--}17 \text{ cm/yr}$ respectively. Second, volcanoes that only deform during the eruptions, like Chaitén and Calbuco. For the first group, the post-eruptive uplift was probably triggered by the large pressure gradient that results from the pressure drop during the Plinian eruptions fed from shallow source (5–10 km depth) and deeper lower crust/mantle sources. This results in at least 0.5 and 0.28 km^3 of basaltic magmas injected during 1993–2025 for Hudson and 2012–2025 for Cordón Caulle respectively. These mechanisms are non-unique. For Cordón Caulle the large pressure gradient triggered by the eruption can also produce post-eruptive uplift driven by fluid remobilization from the bottom to the top of the crystal mush that likely underlies the volcano (Phelps et al., 2023). Eruptions of more than 1 km^3 of basaltic magmas can induce viscoelastic relaxation in addition to magma injection (Li et al., 2021). Considering the complete time interval, Hudson is one of the longest deforming volcanoes observed from InSAR. To our knowledge, only Okmok and Westdahl in the

Aleutians exhibit deformation for more than three decades with rates faster than 2 cm/yr (Angarita et al., 2024; Lu & Dzurisin, 2014; Wang et al., 2023). Cordón Caulle is also the only volcano in the SVZ in which we observe a clear pattern of pre-eruptive uplift, co-eruptive subsidence and post-eruptive uplift that can be interpreted as being produced by magma injection and extraction, as predicted by fluid-solid mechanics models (Lu & Dzurisin, 2014; Pinel et al., 2010), and observed with geodetic data at several basaltic volcanoes in hot-spot and rift environments (Bell, Hernandez, et al., 2021; Parks et al., 2025; Sigmundsson et al., 2018). For the second group, there are multiple explanations for the lack of ground deformation before and after the eruptions. These include pre-eruptive intrusions of small magnitude that are detectable by petrological analyses in samples only but not by geodesy, deformation aliased by the low temporal resolution of InSAR, a highly compressible gas phase in the reservoirs that could mute any detectable pressurization, among others (e.g., Biggs et al., 2014; Biggs & Pritchard, 2017).

3.2. Ground Deformation in the SVZ on a Global Context

On a global scale the satellite geodetic observations of the SVZ are unique (1991-present). It is the region with the highest density of VEI 4–5 eruptions and observations of ground deformation during, before and/or after these events (Figure 1). If the Central and Northern Volcanic Zones along the South American subduction zones are included, the number of sub Plinian and Plinian eruptions in South America in the past three decades increases to six due to the 1993 Lascar (northern Chile) and 2002 Reventador (Ecuador) eruptions. However, no ground geodetic data were recorded during these eruptions, SAR data did not detect ground deformation for Lascar (Pritchard & Simons, 2004a), and were not acquired at Reventador. Deformation data were recorded during and before the VEI 4 2021 Soufrière St. Vincent eruption in the Caribbean (Camejo-Harry et al., 2024). No eruption of this size occurred during the time span along the Central America and Cascades arcs. Three large eruptions occurred along the Aleutian arc (Spurr 1992, Okmok 2008, Kasatochi 2008), but deformation data were recorded only during Okmok 2008 (Freymueller & Kaufman, 2010; Lu & Dzurisin, 2014). Multiple events also occurred at Kamchatka and the Kuriles, but only Shiveluch 2023 showed post-eruptive uplift (Figure S13 in Supporting Information S1, Volkova et al., 2024). In SE Asia and Oceania several eruptions occurred (Table 1), but geodetic observations were well recorded only during the Taal 2020 (Bato et al., 2021) and Rabaul 2006 (Garthwaite et al., 2019; Saunders et al., 2023) eruptions.

In summary, on a global scale, satellite geodetic observations of ground deformation before, during and after sub Plinian and Plinian eruptions have been rarely recorded. The volcanoes with the best available data are the four from the SVZ, Okmok (Figure 5a), Soufrière St. Vincent (Figure 5b), Taal and Rabaul. Cyclic ground deformation like that observed in basaltic volcanoes like Sierra Negra and Grímsvötn has only been observed at Okmok (Figure 5), although the time series of Cordón Caulle resembles a case in which ground deformation has not yet spanned a complete cycle. In contrast, the vast majority of the VEI 4–5 eruptions that occurred in the past three decades lack enough SAR data, are not suitable to be studied with InSAR because they do not sustain high interferometric coherence to yield a positive and/or negative detection of ground deformation, or ground geodetic data were not available. Indeed, more than 50% of these VEI 4–5 eruptions occurred in tropical volcanoes (Indonesia, Papua New Guinea) and polar regions (Alaska, Kamchatka) that are challenging to study with InSAR and with ground monitoring.

3.3. Eruption Triggers From the Global Compilation of Deformation Data

Here we describe general trends for the few cases where deformation time series can be likely related to both magma injection and extraction (Cordón Caulle, Okmok, Taal, Soufrière St. Vincent, Rabaul). On a global scale, the compilation shows that the only subduction volcano where a VEI 4 eruption occurred and where InSAR or GNSS data imaged the entire cycle of deformation is Okmok in Alaska during 1997–2008 (Lu et al., 2010). For the rest of the volcanoes (Cordón Caulle, Soufrière St. Vincent, Taal) the observation time span is short, spanning only a fraction of the inferred period of magma accumulation that leads to an eruption, or has not been studied in detail with source models (Rabaul, Saunders et al., 2023). Therefore, we exclude Rabaul from the following analysis.

Cordón Caulle inflated ~0.6 m during 2003–2011, with no clear deformation above the background level of noise in the InSAR data during January–June 2011 (Delgado, 2021; Euillades et al., 2017; Jay et al., 2014), but there is significant uncertainty due to the presence of atmospheric phase delays in the three months before the eruption. However, significant VT seismicity was recorded during March–June 2011, indicative of low-magnitude

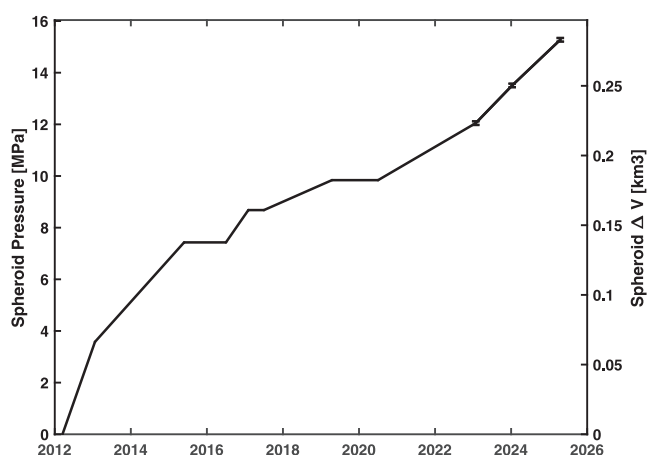


Figure 4. Time series of spheroid volume and pressure change below Cordon Caulle calculated from the joint inversion of InSAR and GNSS data. L-band data became systematically available only after 2021, allowing to estimate data uncertainty in far-field non-deforming regions. Therefore, error bars are shown only after 2022.

pressurization detected by seismology but not by geodesy (Basualto et al., 2023). At Okmok, inflation started right after the VEI 3 1997 eruption, stopped in 2004, and then accelerated in March 2008, although the latter was not well sampled in time by both InSAR and GNSS (Freymueller & Kaufman, 2010; Lu et al., 2010; Figure 5a). Conditions close to tensile failure were reached in the weeks before the eruption (Albright et al., 2019). At Taal, the volcano inflated in several transient episodes that usually lasted less than one year during 1998–2005 recorded by GPS (Galgana et al., 2014), and with InSAR during 2007–2011 (Rivera et al., 2019) and 2015–2020 (Bato et al., 2021). The volcano deflated during 2015–2019 and inflated during 2019–2020, with a total source volume change of 0.045 km³ during the last episode (Bato et al., 2021). At Soufrière St. Vincent geodetic monitoring with GNSS started in 2007, but no pre-eruptive deformation was detected in the years before the eruption, except for uplift six months before the first stage of dome extrusion, with a volume flux of 0.029 km³/yr (Camejo-Harry et al., 2024; Figure 5b).

Ground uplift due to elastic deformation can be recorded for decades without any indication of an eruption, until seismicity increases due to fracturing and inelastic deformation (Robertson & Kilburn, 2016). Several criteria can

explain increases in seismicity and ground deformation before eruptions. The failure forecast method (Kilburn, 2018) indicates that exponential increases in seismicity and ground deformation are a consequence of elastic-brittle failure of the upper crust, resulting from an increase in the strain rate due to fluid movement. This mechanism is conceptually similar to that of Galetto et al. (2022) in which eruptions in basaltic calderas are triggered when the strain rate and magma flux are large and sustained for time scales long enough (0.1 km³/yr and 1 year respectively) to keep the material in an elastic-brittle condition, precluding viscoelastic relaxation. However, these models cannot entirely explain the spectrum of geodetic observations in the studied eruptions. For both Taal 2020 and Soufrière St Vincent 2020–2021 the volume fluxes are at most half the minimum threshold in Galetto et al. (2022) model. At Soufrière St Vincent the abrupt increase in deformation coincided with increases in seismicity (Camejo-Harry et al., 2024), but for Okmok seismicity increased only 5 hr before the 2008 eruption (Roman & Cashman, 2018), despite the uplift that started in March 2008. At Cordón Caulle, deformation slowed or stopped in the months before the 2011–2012 eruption.

There are two common patterns in these volcanoes. First, deformation increased abruptly less than 1 year before the eruptions, except for Cordón Caulle. Second, InSAR and GNSS data only recorded the last episodes of reservoir pressurization from magma injection that brought the stresses around the crystal mush closer to the tensile strength of the rocks until rupture was reached, except for Okmok. As a consequence, the pressure build-up that led to the eruptions must have been episodic and lasting for decades, if not centuries before these eruptions. The exceptionally well recorded November 2023 Grindavik dike intrusion (Iceland) provides further insights to these points. This is clearly not a subduction volcano, but a qualitative comparison can be made for the rupture criteria in the reservoir. A significant part of the pressure build-up that triggered the dike intrusion occurred before of satellite geodetic observations, and was significantly enhanced by tectonics stresses accumulated since the previous eruption in 1280 CE (Sigmundsson et al., 2024). Biggs et al. (2014) have suggested that the lower detection of ground deformation in silicic volcanoes, like most that produce sub Plinian and Plinian eruptions, results from magmas that are rich in volatiles and highly compressible. Therefore, a pulse of injection results in a significant compression of the water-rich magma, leading to no detectable deformation. We consider this a plausible mechanism for the lack of detection in the years before several of these eruptions.

Okmok and Cordón Caulle have the longest records of ground deformation, and we carry out a more detailed modeling to explain the triggering mechanism of their 2008 and 2011 eruptions. For both volcanoes, we assume linear elastic rheologies (e.g., Albright et al., 2019; Delgado et al., 2018), and use a shear modulus of 2.1 GPa, derived from upscaled rock mechanics experiments (Heap et al., 2020). For a spherical chamber, dike propagation is reached when $P = 2T_s$ (Tait et al., 1989), with P the chamber overpressure and T_s the tensile strength of the rock (2–20 MPa, Pinel et al., 2010), so we use pressure as a proxy for T_s . We modeled the Cordón Caulle InSAR data recorded during 2003–2011 (Delgado, 2021; Jay et al., 2014) with the prolate spheroid model of Delgado

et al. (2016), predicting source overpressures of $\Delta P = 0.25$ MPa for 2003–2004, $\Delta P = 0.89$ MPa for 2004–2007, and $\Delta P = 2.6$ MPa for 2007–2008, resulting in $\Delta P = 3.7$ MPa for 2003–2008 (Figure S11 in Supporting Information S1). This is similar to the overpressure found by Novoa et al. (2022), but scaled by the ratio of the shear modulus used by them (20 GPa) with respect to what we used (2.1 GPa). The 2008–2011 deformation results from a pressure change in a different source. During 2008–2011 the volcano rose ~ 15 cm, half that of 2007–2008 (~ 30 cm), and assuming a similar source depth of ~ 5 km (Jay et al., 2014), then the pressure increase during 2008–2011 was at most half (1.3 MPa) compared with that of 2007–2008. Therefore, the 2003–2011 pressure change was at most 5 MPa, or 0.63 MPa/yr. This is lower than the tensile strength of the rock, and as discussed in several studies (Delgado, 2021; Jay et al., 2014), a significant fraction of the mass must have been injected years or decades before 2003 to reach the failure threshold. Using this yearly rate for the inter-eruptive interval between 2011 and the previous eruption in 1960, and neglecting an episode of deflation recorded during 1996–1999 (Pritchard & Simons, 2004a), the total pressure change was 32 MPa. For Okmok, models show that the source is usually very close to symmetry with a radius of 1 km (Albright et al., 2019; Lu & Dzurisin, 2014; Lu et al., 2010). For a spherical chamber model with a total volume change of 0.045 km^3 of magma injected during 1997–2008 inferred from InSAR time series (Lu et al., 2010), the maximum source overpressure is 30 MPa. Then in both volcanoes the tensile strength of the rock was reached after several years of pressure increases. These models are highly non-unique because they do not consider other factors such as magma buoyancy that can also increase the reservoir overpressure (Sigmundsson et al., 2020). This is a feasible mechanism due to the presence of a liquid-rich cap on the top of the crystal mush that likely underlies these volcanoes (Winslow et al., 2022).

The post-eruptive exponential-like uplift at Okmok and Cordón Caulle recorded after their 2008 and 2011 eruptions can be viewed as evidence of the pressure build-up to the next eruption driven by the large co-eruptive pressure gradients. The 13 years of inflation at Cordón Caulle (Figure 3) result in a total of ~ 1.6 m of LOS uplift. We obtain that the spheroid model predicts $\Delta P = 13.5$ MPa (average pressure increase of 1.13 MPa/yr) and a mean magma flux of $0.66 \text{ m}^3/\text{s}$ (Figure 4), for a spheroid of $30\text{--}272 \text{ km}^3$ (Delgado et al., 2019; Novoa et al., 2022). The volume change is similar to that of a sill model of $0.62 \text{ m}^3/\text{s}$ (Delgado et al., 2016). This flux is larger than the long-term flux of Okmok of $0.19 \text{ m}^3/\text{s}$ (Xue et al., 2020), and slightly smaller than the average magma flux at the Laguna del Maule volcanic field of $0.78 \text{ m}^3/\text{s}$ that has inflated ~ 4 m since 2007, but not erupted (Mével et al., 2021). The latter is an extreme example because it has been uplifting with a nearly continuous rate of ~ 25 cm/year between 2007 and 2020, but without any indication that an eruption is about to occur due to the reservoir pressurization. There are few values to compare with other subduction volcanoes, so we compare these fluxes with well-constrained observations in hot spot volcanoes. We note that again, the comparison is not straightforward due to the difference in tectonic regimes, so we focus only on volume fluxes and reservoir dimensions that can be inferred from geodetic data. The Svartsengi reservoir that triggered the Grindavik dike intrusion in the Reykjanes peninsula in Iceland is smaller, 20 km^3 , and its flux of $7.5 \text{ m}^3/\text{s}$ was one order of magnitude larger (Li et al., 2025; Sigmundsson et al., 2024). In contrast, the Cordón Caulle time-averaged magma flux is not large enough for triggering an eruption on time scales of years. Therefore, under the current inferred magma flow of $0.66 \text{ m}^3/\text{s}$ and overpressure of 1.13 MPa/yr it should take a total of 30–35 years to reach again the tensile strength of the rock. For Okmok the situation is different because the intrusion rate after the 2008 eruption is three times larger with respect to the rate during the 1997–2008 pre-eruptive interval (Wang et al., 2021), yet no eruption has occurred so far.

3.4. Long Time Scales of Magma Recharge Before Sub Plinian and Plinian Eruptions in Subduction Volcanoes

Satellite geodetic data suggest that only the last episodes of reservoir pressurization have been detected before the few sub Plinian and Plinian eruptions where observations are available. Clearly, this does not consider the case in which magma was supplied with supply rates that went unnoticed. To better understand the time scales of ground deformation due to magma accumulation derived from geodetic observations, we use a simple dynamic model that couples a pressure source with the mantle through a conduit with a constant pressure in the inlet. Here, the source overpressure ΔP is

$$\Delta P(t) = P(1 - e^{-t/\tau}) \quad (1)$$

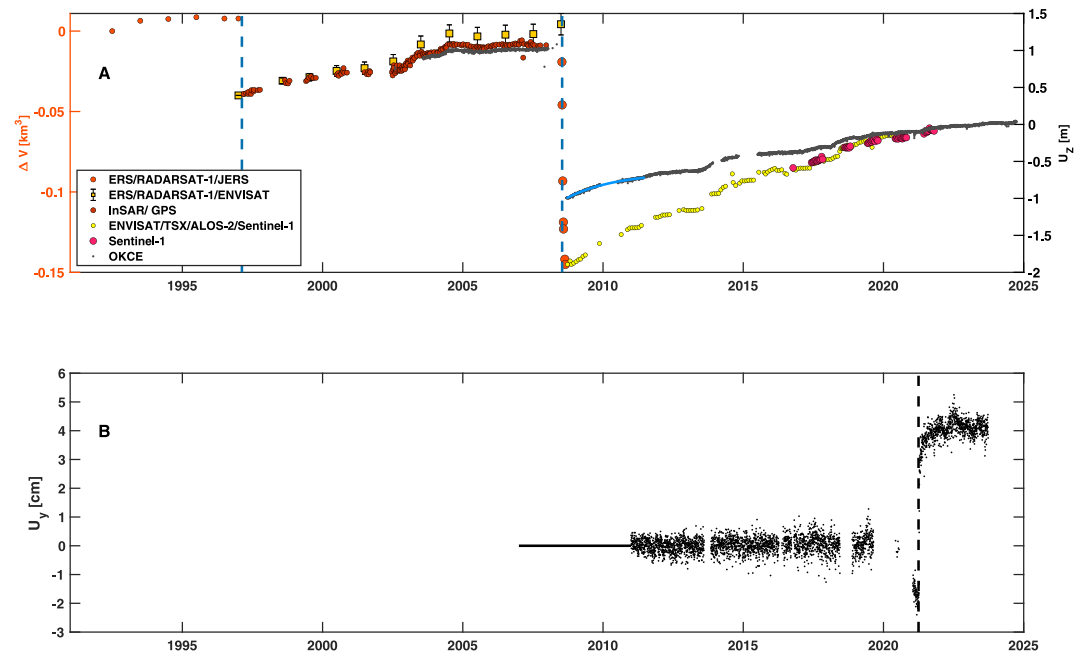


Figure 5. (a) Time series of ground deformation at Okmok caldera spanning the VEI 3 1997 and VEI 4 2008 eruptions. Colored dots are time series of volume change (ΔV) calculated from the inversion of interferograms from different missions during 1992–2020 (indicated in the figure legend, Biggs et al., 2010; Lu et al., 2000, 2005; Lu & Dzurisin, 2010; Xue et al., 2020; Wang et al., 2021). The 2008 co-eruptive deflation is the function $\Delta V = -V_f(1 - e^{-t/\tau})$, with $V_f = 0.14 \text{ km}^3$, $\tau = 13$ days, and interpolated to the dates of the co-eruptive SAR images analyzed by Lu and Dzurisin (2010). Black dots are the vertical displacement (U_z) at GNSS station OKCE in the middle of the caldera spanning 2002 to 2024. The joint time series shows co-eruptive deflation during the 1997 eruption, pre-eruptive source inflation until 2004, slight oscillations including an episode of deflation during 2004–2008 (Caricchi et al., 2014), pre-eruptive uplift in early 2008, co-eruptive subsidence, and post-eruptive uplift with a decreasing exponential trend superimposed atop of exponential trends that occur over shorter time scales (light blue line for 2008–2011, Xue et al., 2020). The GNSS data were manually shifted to lie atop the volume change time series during 2002–2008. (b) Time series of northward displacement (U_y) for the SVGB station in Soufrière St. Vincent volcano with respect to the Caribbean plate. The black horizontal line indicates a period with recorded data (Camejo-Harry et al., 2024), but not available in the time series. The black dashed line is the 2021 VEI 4 eruption. The data clearly records a southward displacement during 2020 due to pre-eruptive inflation and northward displacement due to co-eruptive deflation.

with $P = \Delta P_i + (\rho_r - \rho_m)gL$, ΔP_i the initial magma overpressure, ρ_r and ρ_m wall rock and magma density, and g the gravity acceleration. The time constant is given by $\tau = \frac{8\eta LV(\beta_m + \beta_c)}{\pi R^4}$, with η the magma viscosity, L the conduit length, V the reservoir volume, β_m the magma compressibility, β_c the reservoir compressibility, and R the conduit radius (Lengline et al., 2008; Mével et al., 2016). Basaltic volcanoes in hot spots and rift zones have reservoir volumes between 1 and 21 km^3 for well-monitored eruptions (Anderson et al., 2019; Li et al., 2025; Zheng et al., 2022). In contrast, reservoir volumes of silicic volcanoes are not as nearly as well constrained. For Cordon Caulle the reservoir reaches a volume of $30\text{--}272 \text{ km}^3$ (Delgado et al., 2019; Novoa et al., 2022), one to two orders of magnitude larger. This difference in volume could increase the magnitude of the exponential time constant. We test this hypothesis by considering a shallow storage region at an average depth of 7.4 km, equivalent to the optimal storage pressure of $2 \pm 0.5 \text{ Kbar}$ for subduction volcanoes (Huber et al., 2019), and flow of a gas-poor basaltic magma ($\beta_m = 2 - 4 \times 10^{-10} \text{ Pa}^{-10}$) (Segall, 2010; Zheng et al., 2022) from a deeper region of magma genesis near the crust-mantle interface at an average depth of 33 km. Increasing the reservoir size increases the exponential time scale from 0.4 to 22 years. This results in a very smooth exponential that resembles a linear trend when poorly sampled in time or sampled only during a short time span, for example, the ~ 11 years of Sentinel-1 and ALOS-2 time series (Figure 6). Therefore, one non-unique explanation for the longer time scales in geodetic time series recorded in volcanoes in subduction zones that erupted with VEI 4–5 eruptions compared with those in hot spots is due to the larger sizes of the storage regions. Obviously, this simplistic fluid-solid mechanics model does not consider more complex phenomena like non-linear fluid-solid coupling (Walwer et al., 2022), complex rheologies or the role of regional tectonics (e.g., Sigmundsson et al., 2024), or geophysical

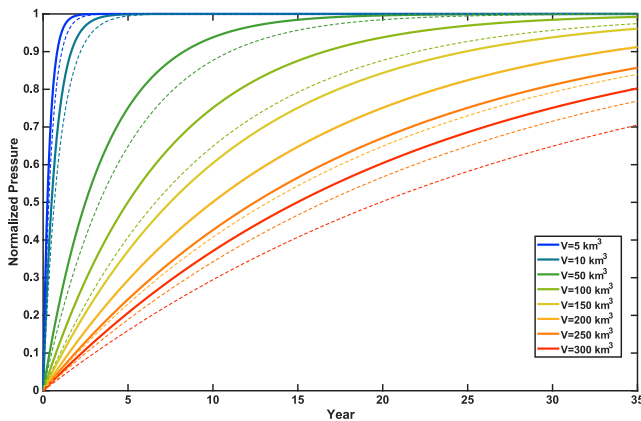


Figure 6. Predicted normalized pressure change as a function of time (Equation 1) for different reservoir sizes from 5 to 300 km³ and for magma compressibilities with $\beta_m = 2 \times 10^{-10}$ Pa⁻¹ (solid lines) and $\beta_m = 4 \times 10^{-10}$ Pa⁻¹ (dashed lines).

images of plumbing systems. In contrast, the 1 year of pre-eruptive increase in the deformation recorded at Taal and Soufrière St. Vincent resembles an increasing exponential trend, and that functional form can be explained as resulting from an acceleration in the magma pressure in the deep mantle source instead of a constant pressure (Mével et al., 2016). This could also be valid for the Okmok eruption, but geodetic data were not well recorded to resolve the temporal evolution of the last year of pre-eruptive ground deformation.

Over time scales of decades, the surface response of an episode of magma injection also depends on the duration of the injection with respect to the viscoelastic relaxation timescales. This leads to less frequent, but larger eruptions (Degruyter & Huber, 2014; Townsend, 2022; Townsend et al., 2019). For a viscoelastic shell surrounding a reservoir, the Maxwell relaxation time is $\tau = \eta/G$, with η the medium viscosity (in this case a shell, Delgado et al., 2018) and G the shear modulus (Segall, 2010). Using a range of values for $\eta = 10^{16}$ – 10^{18} Pa s and $G = 2.1$ GPa we obtain $\tau = 0.15$ – 15 years. This effect should not be important for large strain rates resulting from transient pulses like those in the year before the Taal, Soufrière St.

Vincent, or Okmok eruptions, but should be taken into account for volcanoes that deform over very long time scales. Delgado et al. (2018) found that viscoelastic effects are negligible for Cordón Caulle based on the worst fit to the data of the viscoelastic models compared to the elastic model of magma injection. In contrast, viscoelasticity could be important for Hudson because the volcano deforms with constant rates during the past 30 years, implying either a hydraulic connection with a very long exponential time constant due to a large storage region, that some viscoelastic relaxation occurs, or other phenomena. However, for Hudson, these mechanisms are difficult to differentiate due to the continuous uplift with nearly constant rates and the absence of other monitoring data. Dynamic models that consider poroelasticity and other effects (e.g., Alshembari et al., 2024; Mullet & Segall, 2022; Phelps et al., 2023) should be taken into account in future studies to properly understand the conditions that lead to transient pressure changes and certain cyclic behaviors that also occur during episodes of ground deformation (Walwer et al., 2022).

4. Conclusions

Although more Plinian and sub Plinian eruptions have occurred in subduction zones than well-monitored basaltic eruptions in hot-spots, there is a lack of detailed observations of ground deformation in subduction zones that allow to put bounds on the triggering mechanisms of these eruptions. Indeed, the vast majority of them occurred in volcanoes that were poorly monitored even with InSAR observations that have been systematically acquired all over the Earth since 1991. With this sampling bias, we found three major conclusions. First, ground deformation data with a high temporal resolution have only been acquired before the 2008 Okmok, 2011–2012 Cordón Caulle, 2020 Taal and 2021 Soufrière St. Vincent eruptions. In these four cases, geodesy likely detected the last pulse of ground deformation resulting from a protracted reservoir growth over time scales much longer than those to be recorded by geodesy. Second, only Okmok and Cordón Caulle display the characteristic pattern of pre-eruptive uplift, co-eruptive subsidence, and post-eruptive uplift due to magma injection and extraction observed in basaltic volcanoes in hot-spots and rifts. Third, we speculate that the accumulation is transient, and lasts for much longer times than those recorded by geodesy, and one non-unique explanation could be the large size of reservoirs beneath these volcanoes. The rest of the Plinian and sub-Plinian eruptions in other arcs lack enough data and/or the volcanoes where they occurred do not sustain interferometric coherence to yield a positive and/or negative detection of ground deformation with InSAR, in large part because many of those volcanoes are in tropical areas, small islands or underwater. We speculate that the lack of detection could be a data sampling issue instead of a real geophysical process. Therefore, the lack of an overall trend in the compilation calls for the acquisition of denser time series that might be able to detect deformation of smaller amplitude (<1 cm) in the days before an eruption by new L-band SAR missions such as ALOS-4 or NISAR.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Sentinel-1 data are open access and available at the Alaska Satellite Facility. ALOS-2 ScanSAR data are open access and available at JAXA G-Portal. SAOCOM-1 data are open access and available from CONAE. RADARSAT-2 data were provided by MDA Space under a SOAR-EI project. ALOS-2 stripmap data were provided by JAXA under proposals ER2A2N033, ER3A2N069 and ER4A2N060. These last two data are available after submitting a research proposal to MDA Space and JAXA. The rest of the data are taken from previous studies (Delgado, 2021; Delgado et al., 2014, 2017, 2022; Pritchard & Simons, 2004a). GNSS data for the OKCE and SVGB stations are open access and available from the Nevada Geodetic Laboratory. GNSS time series for Cordón Caulle and Calbuco are available at Zenodo (<https://zenodo.org/records/17780508>). We processed all the InSAR data with the ISCE2 (Rosen et al., 2012) and MintPy software (Yunjun et al., 2019).

Acknowledgments

F.D. thanks Tara Shreve, Raphaël Grandin, Matthew Pritchard, Federico Galetto, Jeffrey Freymueller, Gerald Galgana, Sarah Oliva for discussions, and Zhong Lu for providing the Okmok InSAR time series. We thank detailed reviews by Paul Lundgren and Ronni Grapenthin, and Editors Susanna Ebmeier and Yosuke Aoki for handling the manuscript. This study was funded by FONDECYT de Iniciación 11220513 and FONDECYT Regular 1251171 research Grants (ANID, Chile). JCB would like to thank for the support of the FONDECYT Regular project N°1240501 (ANID, Chile). We thank Observatorio Volcanico de los Andes del Sur (OVDAS) and Centro Sismológico Nacional (CSN) for providing the GNSS time series for Cordón Caulle and Calbuco volcanoes.

References

- Acocella, V. (2021). Volcanoes along convergent plate boundaries. In *Advances in volcanology* (pp. 449–493). https://doi.org/10.1007/978-3-030-65968-4_12/FIGURES/24
- Acocella, V., Ripepe, M., Rivalta, E., Peltier, A., Galetto, F., & Joseph, E. (2023). Towards scientific forecasting of magmatic eruptions. *Nature Reviews Earth & Environment*, 5(1), 5–22. <https://doi.org/10.1038/s43017-023-00492-z>
- Albright, J. A., Gregg, P. M., Lu, Z., & Freymueller, J. T. (2019). Hindcasting magma reservoir stability preceding the 2008 eruption of Okmok, Alaska. *Geophysical Research Letters*, 46(15), 8801–8808. <https://doi.org/10.1029/2019GL083395>
- Alshembari, R., Hickey, J., Pascal, K., & Syers, R. (2024). Declining magma supply to a poroelastic magma mush explains long-term deformation at Soufrière Hills Volcano. *Earth and Planetary Science Letters*, 631, 118624. <https://doi.org/10.1016/j.epsl.2024.118624>
- Anderson, K. R., Johanson, I. A., Patrick, M. R., Gu, M., Segall, P., Poland, M. P., et al. (2019). Magma reservoir failure and the onset of caldera collapse at Kilauea Volcano in 2018. *Science*, 366(6470), eaaz1822. <https://doi.org/10.1126/science.aaz1822>
- Anderson, K. R., Shea, T., Lynn, K. J., Montgomery-Brown, E. K., Swanson, D. A., Patrick, M. R., et al. (2023). The 2018 eruption of Kilauea: Insights, puzzles, and opportunities for volcano science. *Annual Review of Earth and Planetary Sciences*, 52(1), 21–59. <https://doi.org/10.1146/ANNUREV-EARTH-031621-075925>
- Angarita, M., Grapenthin, R., Henderson, S., Christoffersen, M., & Anderson, K. R. (2024). Versatile modeling of deformation (VMOD) inversion framework: Application to 20 years of observations at Westdahl Volcano and fisher caldera, Alaska, US. *Geochemistry, Geophysics, Geosystems*, 25(4), e2023GC011341. <https://doi.org/10.1029/2023GC011341>
- Annen, C., Pinel, V., & Burgisser, A. (2023). Re-evaluation of the role of volatiles in the rupture of magma chambers and the triggering of crystal-rich eruptions. *Journal of Volcanology and Geothermal Research*, 435, 107755. <https://doi.org/10.1016/j.jvolgeores.2023.107755>
- Arzilli, F., Morgavi, D., Petrelli, M., Polacci, M., Burton, M., Genova, D. D., et al. (2019). The unexpected explosive sub-Plinian eruption of Calbuco volcano (22–23 April 2015; southern Chile): Triggering mechanism implications. *Journal of Volcanology and Geothermal Research*, 378, 35–50. <https://doi.org/10.1016/j.jvolgeores.2019.04.006>
- Basualto, D., Tassara, A., Lazo-Gil, J., Franco-Marin, L., Cardona, C., Martín, J. S., et al. (2023). Anatomy of a high-silica eruption as observed by a local seismic network: The June 2011 Puyehue-Cordón Caulle event (southern Andes, Chile). *Solid Earth*, 14(1), 69–87. <https://doi.org/10.5194/SE-14-69-2023>
- Bato, M. G., Lundgren, P., Pinel, V., Solidum, R., Jr., Daag, A., & Cahulogan, M. (2021). The 2020 eruption and large lateral dike emplacement at Taal Volcano, Philippines: Insights from satellite radar data. *Geophysical Research Letters*, 48(7), e2021GL092803. <https://doi.org/10.1029/2021GL092803>
- Bell, A. F., Femina, P. C. L., Ruiz, M., Amelung, F., Bagnardi, M., Bean, C. J., et al. (2021). Caldera resurgence during the 2018 eruption of Sierra Negra volcano, Galápagos Islands. *Nature Communications*, 12(1), 1397. <https://doi.org/10.1038/s41467-021-21596-4>
- Bell, A. F., Hernandez, S., Femina, P. C. L., & Ruiz, M. C. (2021). Uplift and seismicity driven by magmatic inflation at Sierra Negra volcano, Galápagos Islands. *Journal of Geophysical Research: Solid Earth*. <https://doi.org/10.1029/2021JB022244>
- Biggs, J., Ebmeier, S. K., Aspinall, W. P., Lu, Z., Pritchard, M. E., Sparks, R. S. J., & Mather, T. A. (2014). Global link between deformation and volcanic eruption quantified by satellite imagery. *Nature Communications*, 5(1), 3471. <https://doi.org/10.1038/ncomms4471>
- Biggs, J., Lu, Z., Fournier, T., & Freymueller, J. T. (2010). Magma flux at Okmok volcano, Alaska, from a joint inversion of continuous GPS, campaign GPS, and interferometric synthetic aperture radar. *Journal of Geophysical Research: Solid Earth*, 115(B12). <https://doi.org/10.1029/2010jb007577>
- Biggs, J., & Pritchard, M. E. (2017). Global volcano monitoring: What does it mean when volcanoes deform? *Elements*, 13(1), 17–22. <https://doi.org/10.2113/gselements.13.1.17>
- Biggs, J., Robertson, E., & Cashman, K. (2016). The lateral extent of volcanic interactions during unrest and eruption. *Nature Geoscience*, 9(4), 308–311. <https://doi.org/10.1038/ngeo2658>
- Camejo-Harry, M., Pascal, K., Euillades, P., Grandin, R., Hamling, I., Euillades, L., et al. (2024). Monitoring volcano deformation at la Soufrière, St. Vincent during the 2020–21 eruption with insights into its magma plumbing system architecture. *Geological Society, London, Special Publications*, 539(1), 41–62. <https://doi.org/10.1144/SP539-2022-270>
- Caricchi, L., Biggs, J., Annen, C., & Ebmeier, S. (2014). The influence of cooling, crystallisation and re-melting on the interpretation of geodetic signals in volcanic systems. *Earth and Planetary Science Letters*, 388, 166–174. <https://doi.org/10.1016/j.epsl.2013.12.002>
- Caricchi, L., Townsend, M., Rivalta, E., & Namiki, A. (2021). The build-up and triggers of volcanic eruptions. *Nature Reviews Earth & Environment*, 2(7), 458–476. <https://doi.org/10.1038/s43017-021-00174-8>
- Cassidy, M., Ebmeier, S. K., Helo, C., Watt, S. F., Caudron, C., Odell, A., et al. (2019). Explosive eruptions with little warning: Experimental petrology and volcano monitoring observations from the 2014 eruption of Kelud, Indonesia. *Geochemistry, Geophysics, Geosystems*, 20(8), 4218–4247. <https://doi.org/10.1029/2018GC008161>

- Castro, J. M., Schipper, C. I., Mueller, S. P., Militzer, A. S., Amigo, A., Parejas, C. S., & Jacob, D. (2013). Storage and eruption of near-liquidus rhyolite magma at cordon Caulle, Chile. *Bulletin of Volcanology*, 75(4), 17. <https://doi.org/10.1007/s00445-013-0702-9>
- Castruccio, A., Clavero, J., Segura, A., Samaniego, P., Roche, O., Penne, J.-L. L., & Drogue, B. (2016). Eruptive parameters and dynamics of the April 2015 sub-Plinian eruptions of Calbuco Volcano. *Bulletin of Volcanology*, 78(9), 62. <https://doi.org/10.1007/s00445-016-1058-8>
- Chaussard, E., Amelung, F., & Aoki, Y. (2013). Characterization of open and closed volcanic systems in Indonesia and Mexico using InSAR time series. *Journal of Geophysical Research: Solid Earth*, 118(8), 3957–3969. <https://doi.org/10.1002/jgrb.50288>
- Degruyter, W., & Huber, C. (2014). A model for eruption frequency of upper crustal silicic magma chambers. *Earth and Planetary Science Letters*, 403, 117–130. <https://doi.org/10.1016/j.epsl.2014.06.047>
- Delgado, F. (2021). Rhyolitic volcano dynamics in the southern Andes: Contributions from 17 years of InSAR observations at Cordón Caulle volcano from 2003 to 2020. *Journal of South American Earth Sciences*, 106, 102841. <https://doi.org/10.1016/j.jsames.2020.102841>
- Delgado, F., Contreras-Arratia, R., & Samsonov, S. (2022). Magma buoyancy drives rhyolitic eruptions: A tale from the VEI 5 2008–2009 Chaitén eruption (Chile) from seismological and geodetic data. *Earth and Planetary Science Letters*, 590, 117564. <https://doi.org/10.1016/j.epsl.2022.117564>
- Delgado, F., Kubanek, J., Anderson, K., Lundgren, P., & Pritchard, M. (2019). Physicochemical models of effusive rhyolitic eruptions constrained with InSAR and DEM data: A case study of the 2011–2012 Cordón Caulle eruption. *Earth and Planetary Science Letters*, 524, 115736. <https://doi.org/10.1016/j.epsl.2019.115736>
- Delgado, F., Pritchard, M., Ebemier, S., Gonzalez, P., & Lara, L. (2017). Recent unrest (2002–2015) imaged by space geodesy at the highest risk Chilean volcanoes: Villarrica, Llaima, and Calbuco (Southern Andes). *Journal of Volcanology and Geothermal Research*, 344, 270–288. <https://doi.org/10.1016/j.jvolgeores.2017.05.020>
- Delgado, F., Pritchard, M., Lohman, R., & Naranjo, J. A. (2014). The 2011 Hudson volcano eruption (southern Andes, Chile): Pre-eruptive inflation and hotspots observed with InSAR and thermal imagery. *Bulletin of Volcanology*, 76(5), 19. <https://doi.org/10.1007/s00445-014-0815-9>
- Delgado, F., Pritchard, M., Samsonov, S., & Córdova, L. (2018). Renewed post-eruptive uplift following the 2011–2012 rhyolitic eruption of Cordón Caulle (Southern Andes, Chile): Evidence for transient episodes of magma reservoir recharge during 2012–2018. *Journal of Geophysical Research: Solid Earth*, 123(11), 9407–9429. <https://doi.org/10.1029/2018JB016240>
- Delgado, F., Pritchard, M. E., Basualto, D., Lazo, J., Córdova, L., & Lara, L. E. (2016). Rapid reinflation following the 2011–2012 rhyolite eruption at cordon Caulle Volcano (Southern Andes) imaged by InSAR: Evidence for magma reservoir refill. *Geophysical Research Letters*, 43(18), 9552–9562. <https://doi.org/10.1002/2016GL070066>
- Delgado, F., Shreve, T., Borgstrom, S., León-Ibanez, P., Castillo, J., & Poland, M. (2024). A global assessment of SAOCOM-1 L-band stripmap data for InSAR characterization of volcanic, tectonic, cryospheric, and anthropogenic deformation. *IEEE Transactions on Geoscience and Remote Sensing*, 62, 1–21. <https://doi.org/10.1109/TGRS.2024.3423792>
- Eichelberger, J. C. (1995). Silicic volcanism: Ascent of viscous magmas from crustal reservoirs. *Annual Review of Earth and Planetary Sciences*, 23(1), 41–63. <https://doi.org/10.1146/annurev.earth.23.050195.000353>
- Euillades, P. A., Euillades, L. D., Blanco, M. H., Velez, M. L., Grosse, P., & Sosa, G. J. (2017). Co-eruptive subsidence and post-eruptive uplift associated with the 2011–2012 eruption of Puyehue-Cordón Caulle, Chile, revealed by DInSAR. *Journal of Volcanology and Geothermal Research*, 344, 257–269. <https://doi.org/10.1016/j.jvolgeores.2017.06.023>
- Ewert, J., Lockart, A., Marcial, S., & Ambubuyog, G. (1996). *Ground deformation prior to the 1991 eruptions of Mount Pinatubo* (pp. 329–338). PHIVOLCS, University of Washington Press.
- Feuillet, N., Jorry, S., Crawford, W. C., Deplus, C., Thion, I., Jacques, E., et al. (2021). Birth of a large volcanic edifice offshore Mayotte via lithosphere-scale dyke intrusion. *Nature Geoscience*, 14(10), 787–795. <https://doi.org/10.1038/s41561-021-00809-x>
- Freymueller, J. T., & Kaufman, A. M. (2010). Changes in the magma system during the 2008 eruption of Okmok Volcano, Alaska, based on GPS measurements. *Journal of Geophysical Research*, 115(B12), 12415. <https://doi.org/10.1029/2010JB007716>
- Galetto, F., Acocella, V., Hooper, A., & Bagnardi, M. (2022). Eruption at basaltic calderas forecast by magma flow rate. *Nature Geoscience*, 2022(7), 1–5. <https://doi.org/10.1038/s41561-022-00960-z>
- Galetto, F., Pritchard, M. E., Hornby, A. J., Gazel, E., & Mahowald, N. M. (2023). Spatial and temporal quantification of subaerial volcanism from 1980 to 2019: Solid products, masses, and average eruptive rates. *Reviews of Geophysics*, 61(1), e2022RG000783. <https://doi.org/10.1029/2022RG000783>
- Galetto, F., Reale, D., Coppola, D., Sansosti, E., & Pritchard, M. E. (2025). The application of remote sensing data (SAR, thermal and optical) and geodetic modeling to investigate the volcanic activity at Semeru volcano (Indonesia). *Journal of Geophysical Research: Solid Earth*, 130(9), e2025JB031428. <https://doi.org/10.1029/2025JB031428>
- Galgana, G. A., Newman, A. V., Hamburger, M. W., & Solidum, R. U. (2014). Geodetic observations and modeling of time-varying deformation at Taal Volcano, Philippines. *Journal of Volcanology and Geothermal Research*, 271, 11–23. <https://doi.org/10.1016/j.jvolgeores.2013.11.005>
- Garthwaite, M. C., Miller, V. L., Saunders, S., Parks, M. M., Hu, G., & Parker, A. L. (2019). A simplified approach to operational InSAR monitoring of volcano deformation in low-and middle-income countries: Case study of Rabaul Caldera, Papua New Guinea. *Frontiers in Earth Science*, 6, 240. <https://doi.org/10.3389/feart.2018.00240>
- Goitom, B., Oppenheimer, C., Hammond, J. O., Grandin, R., Barrie, T., Donovan, A., et al. (2015). First recorded eruption of Nabro volcano, Eritrea, 2011. *Bulletin of Volcanology*, 77(10), 85. <https://doi.org/10.1007/s00445-015-0966-3>
- Gonnermann, H. M. (2015). Magma fragmentation. *Annual Review of Earth and Planetary Sciences*, 43(43), 431–458. <https://doi.org/10.1146/annurev-earth-060614-105206>
- Gregg, P. M., Zhan, Y., Amelung, F., Geist, D., Mothes, P., Koric, S., & Yunjun, Z. (2022). Forecasting mechanical failure and the 26 June 2018 eruption of Sierra Negra volcano, Galápagos, Ecuador. *Science Advances*, 8(22), 4261. <https://doi.org/10.1126/SCIADV.ABM4261>
- Gudmundsson, M. T., Jonsdottir, K., Hooper, A., Holohan, E. P., Halldorsson, S. A., Ofeigsson, B. G., et al. (2016). Gradual caldera collapse at Bardarbunga volcano, Iceland, regulated by lateral magma outflow. *Science*, 353(6296), 262. <https://doi.org/10.1126/science.aaf8988>
- Hamlyn, J. E., Keir, D., Wright, T. J., Neuberg, J. W., Goitom, B., Hammond, J. O., et al. (2014). Seismicity and subsidence following the 2011 Nabro eruption, Eritrea: Insights into the plumbing system of an off-rift volcano. *Journal of Geophysical Research: Solid Earth*, 119(11), 8267–8282. <https://doi.org/10.1002/2014JB011395>
- Heap, M. J., Villeneuve, M., Albino, F., Farquharson, J. I., Brothelande, E., Amelung, F., et al. (2020). Towards more realistic values of elastic moduli for volcano modelling. *Journal of Volcanology and Geothermal Research*, 390, 106684. <https://doi.org/10.1016/j.jvolgeores.2019.106684>
- Hidayati, S., Triastuty, H., Mulyana, I., Adi, S., Ishihara, K., Basuki, A., et al. (2019). Differences in the seismicity preceding the 2007 and 2014 eruptions of Kelud volcano, Indonesia. *Journal of Volcanology and Geothermal Research*, 382, 50–67. <https://doi.org/10.1016/J.JVOLGEORES.2018.10.017>

- Hreinsdóttir, S. (2019). Volcanoes in Iceland and crustal deformation processes. In *Encyclopedia of complexity and systems science* (pp. 1–22). https://doi.org/10.1007/978-3-642-27737-5_598-2
- Huber, C., Townsend, M., Degruyter, W., & Bachmann, O. (2019). Optimal depth of subvolcanic magma chamber growth controlled by volatiles and crust rheology. *Nature Geoscience*, 12(9), 762–768. <https://doi.org/10.1038/s41561-019-0415-6>
- Jay, J., Costa, F., Pritchard, M., Lara, L., Singer, B., & Herrin, J. (2014). Locating magma reservoirs using InSAR and petrology before and during the 2011–2012 cordon Caulle silicic eruption. *Earth and Planetary Science Letters*, 403, 463. <https://doi.org/10.1016/j.epsl.2014.07.021>
- Kilbride, B. M. C., Edmonds, M., & Biggs, J. (2016). Observing eruptions of gas-rich compressible magmas from space. *Nature Communications*, 7(1), 13744. <https://doi.org/10.1038/ncomms13744>
- Kilburn, C. R. (2018). Forecasting volcanic eruptions: Beyond the failure forecast method. *Frontiers in Earth Science*, 6, 133. <https://doi.org/10.3389/feart.2018.00133>
- Kratzmann, D. J., Carey, S., Scasso, R., & Naranjo, J.-A. (2009). Compositional variations and magma mixing in the 1991 eruptions of Hudson volcano, Chile. *Bulletin of Volcanology*, 71(4), 419–439. <https://doi.org/10.1007/s00445-008-0234-x>
- Kriswati, E., Meilano, I., Hasib, M., Saepuloh, A., Kuncoro, H., Dewanto, B. G., & Fuadi, A. (2024). Explosion mechanism and volume estimation of volcanic ash during the eruption of Sinabung volcano on February 19, 2018: Insight from kinematic GPS and seismic data. *Journal of Volcanology and Geothermal Research*, 447, 108034. <https://doi.org/10.1016/J.JVOLGEORES.2024.108034>
- Lengline, O., Marsan, D., Got, J. L., Pinel, V., Ferrazzini, V., & Okubo, P. G. (2008). Seismicity and deformation induced by magma accumulation at three basaltic volcanoes. *Journal of Geophysical Research: Solid Earth*, 113(B12), 12. <https://doi.org/10.1029/2008jb005937>
- Li, J., Biondi, E., Heimisson, E. R., Puel, S., Zhai, Q., Zhang, S., et al. (2025). Minute-scale dynamics of recurrent dike intrusions in Iceland with fiber-optic geodesy. *Science*, 388(6752), 1189–1193. <https://doi.org/10.1126/SCIENCE.ADU0225>
- Li, S., Sigmundsson, F., Drouin, V., Parks, M. M., Ófeigsson, B. G., Jónsdóttir, K., et al. (2021). Ground deformation after a caldera collapse: Contributions of magma inflow and viscoelastic response to the 2015–2018 deformation field around Bárðarbunga, Iceland. *Journal of Geophysical Research: Solid Earth*, 126(3), e2020JB020157. <https://doi.org/10.1029/2020JB020157>
- Lu, Z., & Dzurisin, D. (2010). Ground surface deformation patterns, magma supply, and magma storage at Okmok volcano, Alaska, from InSAR analysis: 2. Coeruptive deflation, July–August 2008. *Journal of Geophysical Research: Solid Earth*, 115(B5). <https://doi.org/10.1029/2009jb006970>
- Lu, Z., & Dzurisin, D. (2014). *InSAR imaging of aleutian volcanoes*. Springer-Verlag Berlin Heidelberg. https://doi.org/10.1007/978-3-642-00348-6_6
- Lu, Z., Dzurisin, D., Biggs, J., Wicks, C., & McNutt, S. (2010). Ground surface deformation patterns, magma supply, and magma storage at Okmok volcano, Alaska, from InSAR analysis: 1. Interruption deformation, 1997–2008. *Journal of Geophysical Research: Solid Earth*, 115(B5). <https://doi.org/10.1029/2009jb006969>
- Lu, Z., Mann, D., Freymueller, J. T., & Meyer, D. J. (2000). Synthetic aperture radar interferometry of Okmok volcano, Alaska: Radar observations. *Journal of Geophysical Research*, 105(B5), 10791–10806. <https://doi.org/10.1029/2000jb900034>
- Lu, Z., Masterlark, T., & Dzurisin, D. (2005). Interferometric synthetic aperture radar study of Okmok volcano, Alaska, 1992–2003: Magma supply dynamics and postemplacement lava flow deformation. *Journal of Geophysical Research: Solid Earth*, 110(B2). <https://doi.org/10.1029/2004jb003148>
- Major, J. J., & Lara, L. E. (2013). Overview of Chaiten Volcano, Chile, and its 2008–2009 eruption. *Andean Geology*, 40(2), 196–215. <https://doi.org/10.5027/andgeov40n2-a01>
- McKee, C., Itikarai, I., & Davies, H. (2018). Instrumental volcano surveillance and community awareness in the lead-up to the 1994 eruptions at Rabaul, Papua New Guinea. *Advances in Volcanology*, 205–233. https://doi.org/10.1007/11157_2017_4
- McKee, K., Smith, C. M., Reath, K., Snee, E., Maher, S., Matoza, R. S., et al. (2021a). Evaluating the state-of-the-art in remote volcanic eruption characterization Part I: Raikoke volcano, Kuril Islands. *Journal of Volcanology and Geothermal Research*, 419, 107354. <https://doi.org/10.1016/J.JVOLGEORES.2021.107354>
- McKee, K., Smith, C. M., Reath, K., Snee, E., Maher, S., Matoza, R. S., et al. (2021b). Evaluating the state-of-the-art in remote volcanic eruption characterization Part II: Ulawun volcano, Papua New Guinea. *Journal of Volcanology and Geothermal Research*, 420, 107381. <https://doi.org/10.1016/J.JVOLGEORES.2021.107381>
- Mével, H. L., Córdova, L., Cardona, C., & Feigl, K. L. (2021). Unrest at the Laguna del Maule volcanic field 2005–2020: Renewed acceleration of deformation. *Bulletin of Volcanology*, 83(6), 39. <https://doi.org/10.1007/s00445-021-01457-0>
- Mével, H. L., Gregg, P., & Feigl, K. L. (2016). Magma injection into a long-lived reservoir to explain geodetically measured uplift: Application to the 2007–2014 unrest episode at Laguna del Maule volcanic field, Chile. *Journal of Geophysical Research: Solid Earth*. <https://doi.org/10.1002/2016JB013066>
- Morgado, E., Morgan, D. J., Harvey, J., Ángel Parada, M., Castruccio, A., Brahm, R., et al. (2019). Localised heating and intensive magmatic conditions prior to the 22–23 April 2015 Calbuco volcano eruption (southern Chile). *Bulletin of Volcanology*, 81(4), 24. <https://doi.org/10.1007/s00445-019-1280-2>
- Mullet, B., & Segall, P. (2022). The surface deformation signature of a transcrustal, crystal mush-dominant magma system. *Journal of Geophysical Research: Solid Earth*, 127(5), e2022JB024178. <https://doi.org/10.1029/2022JB024178>
- Neal, C. A., Brantley, S. R., Antolik, L., Babb, J. L., Burgess, M., Calles, K., et al. (2018). The 2018 rift eruption and summit collapse of Kīlauea volcano. *Science*, 363(6425), eaav7046. <https://doi.org/10.1126/science.aav7046>
- Nikkhoo, M., Walter, T. R., Lundgren, P. R., & Prats-Iraola, P. (2016). Compound dislocation models (CDMS) for volcano deformation analyses. *Geophysical Journal International*, 208(2), 877–894. <https://doi.org/10.1093/gji/ggw427>
- Novoa, C., Gerbault, M., Remy, D., Cembrano, J., Lara, L. E., Ruz-Ginovés, J., et al. (2022). The 2011 Cordón Caulle eruption triggered by slip on the Liquiñe-Ofqui fault system. *Earth and Planetary Science Letters*, 583, 117386. <https://doi.org/10.1016/j.epsl.2022.117386>
- Parks, M., Drouin, V., Sigmundsson, F., Hjartardóttir, Á. R., Geirsson, H., Pedersen, G. B., et al. (2025). 2023–2024 inflation-deflation cycles at Svartsengi and repeated dike injections and eruptions at the Sundhnúkur crater row, Reykjanes Peninsula, Iceland. *Earth and Planetary Science Letters*, 658, 119324. <https://doi.org/10.1016/J.EPSL.2025.119324>
- Patrick, M. R., Houghton, B. F., Anderson, K. R., Poland, M. P., Montgomery-Brown, E., Johanson, I., et al. (2020). The cascading origin of the 2018 Kīlauea eruption and implications for future forecasting. *Nature Communications*, 11(1), 1–13. <https://doi.org/10.1038/s41467-020-19190-1>
- Pedersen, R., & Sigmundsson, F. (2004). InSAR based sill model links spatially offset areas of deformation and seismicity for the 1994 unrest episode at Eyjafjallajökull volcano, Iceland. *Geophysical Research Letters*, 31(14). <https://doi.org/10.1029/2004GL020368>
- Pedersen, R., & Sigmundsson, F. (2006). Temporal development of the 1999 intrusive episode in the Eyjafjallajökull volcano, Iceland, derived from InSAR images. *Bulletin of Volcanology*, 68(4), 377–393. <https://doi.org/10.1007/s00445-005-0020-y>

- Phelps, P. R., Gonnermann, H. M., Winslow, H., Ruprecht, P., Pritchard, M. E., Delgado, F., & Liao, Y. (2023). Feasibility of melt segregation from a crystal mush in response to the 2011–12 eruption at Cordón Caulle, Chile. *Geophysical Journal International*, 235(1), 610–634. <https://doi.org/10.1093/GJI/GGAD259>
- Pinel, V., Jaupart, C., & Albino, F. (2010). On the relationship between cycles of eruptive activity and growth of a volcanic edifice. *Journal of Volcanology and Geothermal Research*, 194(4), 150–164. <https://doi.org/10.1016/j.jvolgeores.2010.05.006>
- Pinel, V., Poland, M. P., & Hooper, A. (2014). Volcanology: Lessons learned from synthetic aperture radar imagery. *Journal of Volcanology and Geothermal Research*, 289, 81–113. <https://doi.org/10.1016/j.jvolgeores.2014.10.010>
- Pinel, V., Sigmundsson, F., Sturkell, E., Geirsson, H., Einarsson, P., Gudmundsson, M. T., & Högnadóttir, T. (2007). Discriminating volcano deformation due to magma movements and variable surface loads: Application to Katla subglacial volcano, Iceland. *Geophysical Journal International*, 169(1), 325–338. <https://doi.org/10.1111/j.1365-246x.2006.03267.x>
- Pritchard, M. E., & Simons, M. (2004a). An InSAR-based survey of volcanic deformation in the southern Andes. *Geophysical Research Letters*, 31(15), 4. <https://doi.org/10.1029/2004gl020545>
- Pritchard, M. E., & Simons, M. (2004b). An InSAR-based survey of volcanic deformation in the Central Andes. *Geochemistry, Geophysics, Geosystems*, 5(2), 42. <https://doi.org/10.1029/2003gc000610>
- Reath, K., Pritchard, M., Poland, M., Delgado, F., Carn, S., Coppola, D., et al. (2019). Thermal, deformation, and degassing remote sensing time series (CE 2000–2017) at the 47 most active volcanoes in Latin America: Implications for volcanic systems. *Journal of Geophysical Research: Solid Earth*, 124(1), 195–218. <https://doi.org/10.1029/2018JB016199>
- Ripepe, M., Lacanna, G., Pistolesi, M., Silengo, M. C., Aiuppa, A., Laiolo, M., et al. (2021). Ground deformation reveals the scale-invariant conduit dynamics driving explosive basaltic eruptions. *Nature Communications*, 12(1), 1–8. <https://doi.org/10.1038/s41467-021-21722-2>
- Rivera, A. M. M., Amelung, F., Albino, F., & Gregg, P. M. (2019). Impact of crustal rheology on temperature-dependent viscoelastic models of volcano deformation: Application to Taal Volcano, Philippines. *Journal of Geophysical Research: Solid Earth*, 124(1), 978–994. <https://doi.org/10.1029/2018JB016054>
- Robertson, R. M., & Kilburn, C. R. (2016). Deformation regime and long-term precursors to eruption at large calderas: Rabaul, Papua New Guinea. *Earth and Planetary Science Letters*, 438, 86–94. <https://doi.org/10.1016/j.epsl.2016.01.003>
- Roman, A., & Lundgren, P. (2021). Dynamics of large effusive eruptions driven by caldera collapse. *Nature*, 592(7854), 392–396. <https://doi.org/10.1038/s41586-021-03414-5>
- Roman, A., & Lundgren, P. (2023). Magma transport and storage along rift zones through volcanic conduits. *Journal of Geophysical Research: Solid Earth*, 128(8), e2023JB026509. <https://doi.org/10.1029/2023JB026509>
- Roman, D. C., & Cashman, K. V. (2018). Top-down precursory volcanic seismicity: Implications for ‘stealth’ magma ascent and long-term eruption forecasting. *Frontiers in Earth Science*, 6, 124. <https://doi.org/10.3389/feart.2018.00124>
- Rosen, P. A., Gurrola, E., Sacco, G. F., & Zebker, H. (2012). The InSAR scientific computing environment, EUSAR 2012: 9th European Conference on Synthetic Aperture Radar, pp. 730–733. <https://github.com/isce-framework/isce2>
- Saunders, S., Tenor, E., Wakawa, J., & Nohou, J. (2023). Twenty-two years of GPS monitoring at Rabaul caldera, a narrative history. *Geosciences*, 13(8), 249. <https://doi.org/10.3390/GEOSCIENCES13080249>
- Segall, P. (2010). *Earthquake and volcano deformation* (p. 432). Princeton University Press.
- Segall, P., Anderson, K., & Wang, T. A. (2022). Could Kilauea’s 2020 post caldera-forming eruption have been anticipated? *Geophysical Research Letters*, 49(15), e2022GL099270. <https://doi.org/10.1029/2022GL099270>
- Shreve, T., & Delgado, F. (2023). Trapdoor fault activation: A step toward caldera collapse at Sierra Negra, Galápagos, Ecuador. *Journal of Geophysical Research: Solid Earth*, 128(5), e2023JB026437. <https://doi.org/10.1029/2023JB026437>
- Shreve, T., Grandin, R., & Boichu, M. (2022). Reservoir depressurization driven by passive gas emissions at Ambrym volcano. *Earth and Planetary Science Letters*, 584, 117512. <https://doi.org/10.1016/j.epsl.2022.117512>
- Shreve, T., Grandin, R., Boichu, M., Garaebiti, E., Moussallam, Y., Ballu, V., et al. (2019). From prodigious volcanic degassing to caldera subsidence and quiescence at Ambrym (Vanuatu): The influence of regional tectonics. *Scientific Reports*, 9(1), 18868. <https://doi.org/10.1038/s41598-019-55141-7>
- Sigmundsson, F., Parks, M., Geirsson, H., Hooper, A., Drouin, V., Vogfjörð, K. S., et al. (2024). Fracturing and tectonic stress drives ultrarapid magma flow into dikes. *Science*, 383(6688), 1228–1235. <https://doi.org/10.1126/SCIENCE.ADN2838>
- Sigmundsson, F., Parks, M., Hooper, A., Geirsson, H., Vogfjörð, K. S., Drouin, V., et al. (2022). Deformation and seismicity decline before the 2021 Fagradalsfjall eruption. *Nature*, 609(7927), 523–528. <https://doi.org/10.1038/s41586-022-05083-4>
- Sigmundsson, F., Parks, M., Pedersen, R., Jónsdóttir, K., Ófeigsson, B. G., Grapenthin, R., et al. (2018). *Magma movements in volcanic plumbing systems and their associated ground deformation and seismic patterns* (pp. 285–322). Elsevier. <https://doi.org/10.1016/B978-0-12-809749-6.00011-X>
- Sigmundsson, F., Pinel, V., Grapenthin, R., Hooper, A., Halldórsson, S. A., Einarsson, P., et al. (2020). Unexpected large eruptions from buoyant magma bodies within viscoelastic crust. *Nature Communications*, 11(1), 2403. <https://doi.org/10.1038/s41467-020-16054-6>
- Sigmundsson, F., Pinel, V., Lund, B., Albino, F., Pagli, C., Geirsson, H., & Sturkell, E. (2010). Climate effects on volcanism: Influence on magmatic systems of loading and unloading from ice mass variations, with examples from Iceland. *Philosophical Transactions of the Royal Society A: Mathematical, Physical & Engineering Sciences*, 368(1919), 2519–2534. <https://doi.org/10.1098/rsta.2010.0042>
- Smittarello, D., Smets, B., Barrière, J., Michellier, C., Oth, A., Shreve, T., et al. (2022). Precursor-free eruption triggered by edifice rupture at Nyiragongo volcano. *Nature*, 609(7925), 83–88. <https://doi.org/10.1038/s41586-022-05047-8>
- Surono, P. J., Pallister, J., Boichu, M., Buongiorno, M. F., Budisantoso, A., Costa, F., et al. (2012). The 2010 explosive eruption of Java’s Merapi volcano—A ‘100-year’ event. *Journal of Volcanology and Geothermal Research*, 241–242, 121–135. <https://doi.org/10.1016/J.JVOLGEORES.2012.06.018>
- Tait, S., Jaupart, C., & Vergnolle, S. (1989). Pressure, gas content and eruption periodicity of a shallow, crystallizing magma chamber. *Earth and Planetary Science Letters*, 92(1), 107–123. [https://doi.org/10.1016/0012-821x\(89\)90025-3](https://doi.org/10.1016/0012-821x(89)90025-3)
- Townsend, M. (2022). Linking surface deformation to thermal and mechanical magma chamber processes. *Earth and Planetary Science Letters*, 577, 117272. <https://doi.org/10.1016/J.EPSL.2021.117272>
- Townsend, M., Huber, C., Degruyter, W., & Bachmann, O. (2019). Magma chamber growth during inter-caldera periods: Insights from thermo-mechanical modeling with applications to Laguna del Maule, Campi Flegrei, Santorini, and Aso. *Geochemistry, Geophysics, Geosystems*, 20(3), 1574–1591. <https://doi.org/10.1029/2018GC008103>
- Volkova, M. S., Mikhailov, V. O., & Gorbach, N. V. (2024). The nature of deformations of the western slope of Shiveluch volcano after the eruption on April 11, 2023, identified by SAR interferometry. *Doklady Earth Sciences*, 518(2), 1723–1731. <https://doi.org/10.1134/S1028334X24602694>

- Walwer, D., Ghil, M., & Calais, E. (2022). A data-based minimal model of episodic inflation events at volcanoes. *Frontiers in Earth Science*, 10, 877. <https://doi.org/10.3389/FEART.2022.759475>
- Wang, J., Lu, Z., Bekaert, D., Marshak, C., Govorcin, M., Sangha, S., et al. (2023). Along-arc volcanism in the western and central Aleutian from 2015 to 2021 revealed by cloud-based InSAR processing. *Geophysical Research Letters*, 50(23), e2023GL106323. <https://doi.org/10.1029/2023GL106323>
- Wang, J., Lu, Z., & Gregg, P. M. (2021). Inflation of Okmok volcano during 2008–2020 from PS analyses and source inversion with finite element models. *Journal of Geophysical Research: Solid Earth*, 126(10), e2021JB022420. <https://doi.org/10.1029/2021JB022420>
- Winslow, H., Ruprecht, P., Gonnermann, H. M., Phelps, P. R., Muñoz-Saez, C., Delgado, F., et al. (2022). Insights for crystal mush storage utilizing mafic enclaves from the 2011–12 Cordón Caulle eruption. *Scientific Reports*, 12(1), 1–14. <https://doi.org/10.1038/s41598-022-13305-y>
- Xu, W. B., Jonsson, S., Ruch, J., & Aoki, Y. (2016). The 2015 Wolf volcano (Galapagos) eruption studied using Sentinel-1 and ALOS-2 data. *Geophysical Research Letters*, 43(18), 9573–9580. <https://doi.org/10.1002/2016gl069820>
- Xue, X., Freymueller, J., & Lu, Z. (2020). Modeling the post-eruptive deformation at Okmok based on the GPS and InSAR time series: Changes in the shallow magma storage system. *Journal of Geophysical Research: Solid Earth*, 125(2), e2019JB017801. <https://doi.org/10.1029/2019JB017801>
- Yip, S. T. H., Biggs, J., Edmonds, M., Liggins, P., & Shorttle, O. (2022). Contrasting volcanic deformation in arc and ocean island settings due to exsolution of magmatic water. *Geochemistry, Geophysics, Geosystems*, 23(7), e2022GC010387. <https://doi.org/10.1029/2022GC010387>
- Yunjun, Z., Fattahi, H., Amelung, F., (2019). Small baseline InSAR time series analysis: Unwrapping error correction and noise reduction. *Computers & Geosciences*, 133, 104331. <https://doi.org/10.1016/j.cageo.2019.104331>
- Zheng, Y., Blackstone, L., & Segall, P. (2022). Constraints on absolute magma chamber volume from geodetic measurements of trapdoor faulting at Sierra Negra volcano, Galapagos. *Geophysical Research Letters*, 49(5), e2021GL095683. <https://doi.org/10.1029/2021GL095683>

References From the Supporting Information

- Báez, J. C., Leyton, F., Troncoso, C., Campo, F. D., Bevis, M., Vigny, C., et al. (2018). The Chilean GNSS network: Current status and progress toward early warning applications. *Seismological Research Letters*, 89(4), 1546–1554. <https://doi.org/10.1785/0220180011>
- Chen, C. W., & Zebker, H. A. (2001). Two-dimensional phase unwrapping with use of statistical models for cost functions in nonlinear optimization. *Journal of the Optical Society of America a-Optics Image Science and Vision*, 18(2), 338–351. <https://doi.org/10.1364/josaa.18.000338>
- Fattahi, H., Agram, P., & Simons, M. (2017a). A network-based enhanced spectral diversity approach for tops time-series analysis. *IEEE Transactions on Geoscience and Remote Sensing*, 55(2), 777–786. <https://doi.org/10.1109/TGRS.2016.2614925>
- Fattahi, H., Simons, M., & Agram, P. (2017b). InSAR time-series estimation of the ionospheric phase delay: An extension of the split range-spectrum technique. *IEEE Transactions on Geoscience and Remote Sensing*, 55(10), 5984–5996. <https://doi.org/10.1109/TGRS.2017.2718566>
- Liang, C., & Fielding, E. J. (2017). Interferometry with ALOS-2 full-aperture ScanSAR data. *IEEE Transactions on Geoscience and Remote Sensing*, 55(5), 2739–2750. <https://doi.org/10.1109/TGRS.2017.2653190>
- Lohman, R. B., & Simons, M. (2005). Some thoughts on the use of InSAR data to constrain models of surface deformation: Noise structure and data downsampling. *Geochemistry, Geophysics, Geosystems*, 6(1), 12. <https://doi.org/10.1029/2004gc000841>